

Checklist/Planning

Title page:

- ☐ Title (12 words max, in title case)
- ☐ Name of each author and their institutional affiliations
- ☐ All of our emails
- ☐ Word count w/o figures, tables, references, and the title page

Keywords:

- ☒ 3-5 keywords

Climate Change, Agriculture, Sustainability, Food Insecurity, Predictive Analysis

Abstract:

The increased effect of climate change and natural disaster frequency (probably link a study relating climate change to increased natural disaster?) has impacted agricultural productivity deeply. Around the globe, different countries have addressed climate change through international protocols, accords, and other collaborations through organizations such as the United Nations. As sustainable practices have been introduced and implemented in several different countries, the possibility of maintaining these costly practices may not be viable or widespread in certain countries. Within this analysis, five different countries of different socioeconomic standing are analyzed based on factors such as GDP per capita, arable land, and emissions directly from agriculture over a period of fifty years in order to predict the extent of food insecurity in the next [however many] years. Through studying the individual socioeconomic factors, a more distinct and individualized model is produced. Methods methods methods methods methods. Results results results results. Conclusions that can be made.

- ☐ State aim and scope of research
- ☐ Methods - skip for now until we figure it out
- ☐ Results
- ☐ Conclusions
- ☐ Implications of findings
- ☐ Easy to understand
- ☐ No references or references to figures/tables
- ☐ 250 word limit

Introduction

planning structure???:

- Climate change super big, super impactful,
- In agriculture specifically, climate change obv. is going to impact, and thru a lot of different ways sjdkjadkflfkakjfk → good chance to slot in variables here
- Agri production important → slot in food scarcity, production, econ impacts, idk.
- Therefore must adapt agri production to climate → this is ✨sustainable agriculture ✨
✨woahhhhh

- To know how to sustainable agriculture, must know how climate change this that bam wham make super wonky impacts, thru what ways/avenues and means, and what specifically needs to change in response.
- Goal of study is to look at how much impact climate change on agri production
***FACT CHECK ME ON THIS TvT + the other stuff we're studying cuz i'm kinda sorta lost heh...

☐ Focused research question

☐ Define the variables being studied

☐ Definition of GDP,, emissions, HDI, uhhh what else?

☐

☐ HDI - The Human Development Index is a metric combining average lifespan, education, and standard of living and is often used to assess and compare the development as well as standard of living. → use IHDI if data is also available cuz accounts for inequalities

☐ Sources: <https://ourworldindata.org/human-development-index>

☐ <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

☐ <https://hdr.undp.org/inequality-adjusted-human-development-index#/indicies/IHDI>

☐ GDP per capita - gross domestic product, based on 2015 US dollars for standardization

☐ Food insecurity - ** pretty important wow :0** avg % adequate caloric yk... intake i think? → look at arthur's source thing TwT

☐ Option 1: divide food insecurity into more categories like FAO and connect directly to specific factors or groups of factors

☐ Access to food (FIES) → severity

☐ Access to dietary energy (PoU) → monitor hunger at global level

☐ Food consumption (Household consumption and expenditure) → nutrient consumption/ intake

☐ Option 2: use one singular definition (take from one of three or one larger uniform one) → **ASK CATHYYYY!!!**

☐ Objective of work

☐ Relevance of work -

☐ Importance of work

- Climate change affects us all

- Countries are able to continue to grow economically or begin to do so with practices that relate to their financial capacity 🕶️

- Continuing to develop and improve conditions within countries that rely more on agriculture may benefit the overall conditions of the country

- Sustainable practices also have positive long-term effects, and should, at the least, be more environmentally sound in the long run compared to the status quo

☐ Describe datasets:

- ☐ Origin
- ☐ Data reliability
- ☐ Role in project

Methods - basically look at plan of attack

- ☐ Explain materials and methods in a way that other people can replicate the process
- ☐ Explain methods and reasons for using them
- ☐ Sources for materials used (paragraph form)
- ☐ Methods written in step by step process
 - ☐ Explain how variables were measured
- ☐ Model
 - ☐ Detail step by step process for how the model is created
 - ☐ How model is build on existing frameworks
 - ☐ Explain choices of calculations and statistical analysis techniques

Results:

- ☐ Include ALL data even if it's contrary to our hypothesis
- ☐ Report all observations made during research
 - ☐ Avoid selection bias
 - ☐ Identify all possible outliers
- ☐ no data analysis in this part!!
- ☐ State what can be observed from the data and general trends
- ☐ Original investigations:
 - ☐ Present experimental outcomes in the form of tables and figures
 - ☐ Mainly explain observations, patterns, and trends using scientific theory and reasoning
- ☐ Machine learning: present and contextualize relevant benchmarks

Discussion:

- ☐ Interpretation of the results in relation to original research objective
- ☐ Briefly summarize key findings
- ☐ Comment on significance of said findings
- ☐ Present possible suggestions for future directions of research
- ☐ MUST be in paragraph form
- ☐ Relevance of key findings in relation to other literature in the field
 - ☐ Present other credible sources/academic research to support or contrast our results/interpretations

Conclusion

- ☐ Reiterate primary findings to summarize the main results and discussion points

Acknowledgements

- ☐ Ren's mom

References:

- ☐ External info/data cited where the source is relevant
- ☐ All sources listed in a bibliographic reference section at the end titled References
- ☐ Bibliography must be in Vancouver
- ☐ In text citations must be [between brackets]

Appendices:

- ☐ Data beyond what is included in the result section (optional)
- ☐ Stands separately from the rest of the article

Draft

From Harvest to Food Security: Machine Learning Forecasts for the United States

Cathy Rong, Safiya Taher, Sabrina Qian, Renisa Ghosh, Arthur Jiang

Abstract

Food insecurity remains a persistent challenge, even in high-producing agricultural systems. Timely national-level indicators are needed to support monitoring and policy decisions. This study tests whether U.S. agricultural production signals can predict a composite Food Security Index (FSI) derived from food-system outcomes. We constructed an annual United States dataset spanning from 1970 to 2023 that integrates USDA NASS QuickStats measures for three major staple crops, corn, soybeans, and wheat (yield, production, and area harvested).

The prediction target is an FSI scaled from 0 to 1, where higher values indicate lower food insecurity, constructed through mathematical modeling of four FAOSTAT variables: dietary energy supply, protein supply, GDP per capita, and a food production index. We compared multiple supervised regression models using a fixed split of 1970 to 2012 data for training, and a testing period of 2013 to 2023. We further included a change-based formulation that predicts a year to year change to FSI levels (Δ FSI), with engineered shock and lag features.

A Random Forest model trained on the Δ FSI feature set performed the best, achieving a Root Mean Squared Error (RMSE) of about 0.066 on the 2013-2023 testing data and improving on a persistence baseline. Overall, results showed that the national crop statistics contain measurable predictive signals for annual food security dynamics and motive extensions.

Introduction

Even in highly productive agricultural systems, food insecurity is still a continuous challenge as national food security depends on more than the total output of crops [#]. Beyond aggregated supply, food security is a reflection on whether or not people can consistently access safe, sufficient, and nutritious food while the system remains stable under shocks [#]. As climate variability [#], market disruptions [#], and economic pressures interact, there is an increased interest in timely, interpretable national indicators that can be monitored over long periods in support of decision making. [#]

Agricultural production is one of the most consistently measured national variables. In the United States, corn, soybeans, and wheat are especially well documented as they dominate staple crop production and influence downstream food and feed markets [#] [#]. Yearly changes in yield, acreage, and total production have the potential to capture biophysical constraints and

economic land-use responses. However, directly linking production to national food security outcomes is not straightforward; food security also depends on economic access [#], supply chains [#] [#], and broader macro conditions, which introduce nonlinear relationships and time-lagged effects.

We created a composite food security index (FSI), computed from four different components derived from the Food and Agriculture Organization Corporate Statistical (FAOSTAT) Database: dietary energy supply (kcal/capita/day), protein supply (g/capita/day), GDP per capita, and a food production index, to act as a summary measure score intended to track how “food-secure” the country is each year by using a series of macro indicators that are consistently available over longer time spans. It is a proxy for overall food security conditions rather than a measure for household hunger. Dietary energy supply provides information on food availability in the national supply [#]; protein supply shows a diet quality or adequacy dimension to the mix, considering beyond calories [#]; GDP per capita is used widely as a macro proxy for economic access and affordability where the higher the income, the higher the ability to purchase food, buffer price shocks, etc (generally) [#] ; finally, the food production index is a broad domestic production availability signal by tracking the volume or value of edible, nutritious food crops relative to a base period. [#]

This idea came from reviewing past composite food indices also built by selecting indicators and then normalizing and aggregating them. For example, the Global Food Security Index (GFSI) considers multiple indicators (food affordability, availability, quality and safety, sustainability and adaptation) over 113 different countries. [#] These past indexes inspired us to create our FSI as a defensible, simple, and transparent measure, tailored reliably to what we are trying to solve.

Methods

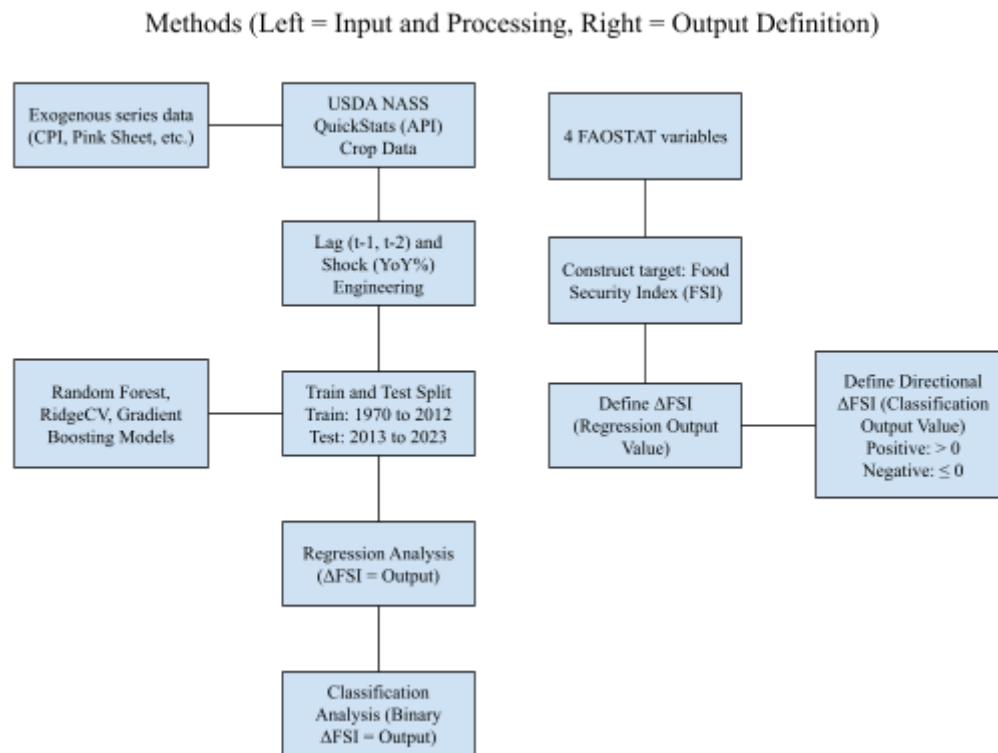
Data and Study Design

We evaluated whether or not national staple crop production can predict a composite FSI for the United States (U.S.) using an annual time series (1970-2023). Three different crop predictors were compiled from the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) Quickstats: corn, soybeans, and wheat, including yield, production, and harvested area for each crop (N=1,944). The prediction target was an FSI scaled to [0, 1], where a higher value signaled better food security).

In addition to crop exclusive statistics, we included a set of exogenous drivers to capture broader conditions that can influence food security and crop performance. U.S. food-price conditions were represented using a food Consumer Price Index (CPI) index and its derived annual change. The World Bank Commodity Price Data (“Pink Sheet”) indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals & Minerals) and selected commodity price series (crude oil, maize, soybeans, and U.S. wheat prices: Hard Red Winter, HRW, and Soft Red Winter, SRW) acted as global input and commodity pressures.

All raw datasets were first restricted to the United States and then converted to annual values and merged by year to form a single master table. Non-numeric values were treated as missing and excluded only when required for the given model fit to maintain a consistent predictor-target alignment. The resulting dataset was then used for feature engineering (lags and year-over-year “shock” variables) and model evaluation under time-respecting train/test splits.

Figure 1 presents the flow chart for the machine learning model, including the data integration and model validation.



Notably, the QuickStats data does not link individual states across years, ensuring these data are applied across the US. To minimize bias and maintain model integrity, no state data were included in the features. This approach ensured a clean separation of training, validation, and testing data to support robust model development and evaluation.

Data Preparation

We used the USDA NASS QuickStats API to extract our crop predictors. Specifically, we sent requests that filtered records to U.S level, annual observations and targeted our three staple commodities (corn, soybeans, and wheat) as well as the required measures (yield, production, and area harvested). Each request was run through a Python code using the library ‘requests’ to call the API and saved each metric into its own structured dataset in CSV form. In addition, we

downloaded exogenous series from the U.S. Bureau of Labor Statistics (CPI), the World Bank Commodity Price Data (“Pink Sheet”) was used for indices and commodity prices and U.S. population totals from the World Bank population series.

Data cleaning and restructuring were also performed in Python, using the ‘pandas’ and ‘NumPy’ libraries for parsing, value conversion, reshaping, merging, and feature engineering (shock and lag value calculation). For consistency across all sources, all non-numeric values were dropped when required for a given model fit. In addition, we also used ‘pathlib’ for file handling, and ‘scikit-learn’ utilities to support model-ready preprocessing in our workflow. All of the cleaned series were then reshaped into a year-indexed table, at one row per year, and merged with the constructed FSI and exogenous predictors to create the final modeling dataset spanning from 1970 to 2023.

Targets and Feature Engineering

We created the FSI in multiple steps. For FSI(t), we defined it as a single number between 0 and 1 for each year t, where the higher the value, the better food security (lower insecurity). We built it off of four annual national indicators:

1. K(t): Dietary energy supply (kcal per capita per day)
2. P(t): Protein supply (g per capita per day)
3. G(t): GDP per capita (USD)
4. F(t): Food production index

Because these four variables have different units and ranges, we first normalized each one to the same [0, 1] scale as the final FSI over our full study cohort of 1970-2023, making them directly comparable. By using min-max normalization:

$$X_{norm}(t) = \frac{X(t) - \min(X)}{\max(X) - \min(X)} \text{ for } X \in \{K, P, G, F\}$$

Each component lies in a range of [0,1] after this.

Secondly, we combined all four components into one index by equal-weighted form. This is because we did not set any weights to any of the components, making all of them of equal importance to one another.

$$FSI(t) = \frac{K_{norm}(t) + P_{norm}(t) + G_{norm}(t) + F_{norm}(t)}{4}$$

This, in turn, makes FSI(t) a value within [0,1], where a higher FSI value corresponds to stronger food security conditions, by our composite definition. At the same time, a drop in FSI would signify a deterioration driven by one or more components declining in relevance to their historical range.

While FSI provides an interpretable annual picture, its strong persistence over time reflects the idea that forecasting will be easier in year-to-year change increments rather than absolute levels. Therefore, we tested two prediction formulations: level and change-based. In the level formulation, models predict FSI(t) directly, where FSI(t) denotes the FSI in year t. In the change based formulation, models predict annual change $\Delta FSI(t) = FSI(t) - FSI(t-1)$ and reconstruct

levels via $FSI'(t) = FSI(t-1) + \Delta FSI'(t)$. Where this year's predicted FSI equals last year's FSI plus the model's predicted change this year. FSI' means that said value is a predicted one. Predictors included crop indicators (yield, production, and area harvested for corn, soybeans, and wheat) as well as exogenous variables in a climate and economic context.

To capture disruptions and delayed effects, we manually created shock features and lags for all time-varying predictors. Shocks were computed on a year by year basis as changes (percentage changes for strictly positive variables) We also included 1 to 2 year lags of both raw and shocked level terms. This design allows our model to learn immediate versus delayed associations between production, external conditions, and changes in national food security.

Model Selection

To accurately test whether or not all of the predictors could contain a usable signal for food security, we employed multiple supervised machine learning regression models to predict for our FSI. The model had a time-respective training and testing data split where 1970 to 2012 data was used for training and 2013 to 2023 data was used for testing. The model was benchmarked against a persistent baseline that forecasts the next year's value.

We compared a small set of regression models to test, each with different strengths. These included a regularized Ridge regression linear model as well as a non linear Random Forest regression ensemble model. The Ridge model provides a transparent benchmark that handles all correlated predictors through regularization; the Random Forest model can help capture non-linear interactions between variables without requiring a separate fixed functional form. To evaluate which model allowed for easier forecasting, our two formulation processes were used, and the best-performing approach was selected based on predictive accuracy and the held-on test period.

Performance Metrics

Predictive accuracy on held-out years was measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 values, reported in relevance to the persistence baseline. For the second formulation, the change-based formulation, we assessed directional performance additionally by classifying whether ΔFSI was bigger than 0 and then computing a confusion matrix and balanced accuracy along with it. Receiver Operating Characteristic Area Under the Curve (ROC AUC) and average precision were also calculated using the changed, predicted scores.

Results

Through national staple-crop statistics and exogenous economic indicators, we evaluated the predictive perforation of the proposed Ridge and Random Forest models for estimating food security in the United States. In this section, we see how well our models predict U.S. food security in the held-out period and what signals drive said predictions.

According to the results across the tested models, the Random Forest model, trained on ΔFSI , was selected as the primary approach as it provided the most reliable out-of-sample tracking of FSI relative to the alternative models and our persistence baseline. Therefore, we present the results from the Random Forest model first, followed by diagnostics and an auxiliary directional evaluation.

Figure 2: Observed vs Predicted FSI Over Time: Random Forest (ΔFSI) vs Persistence (2013–2023)

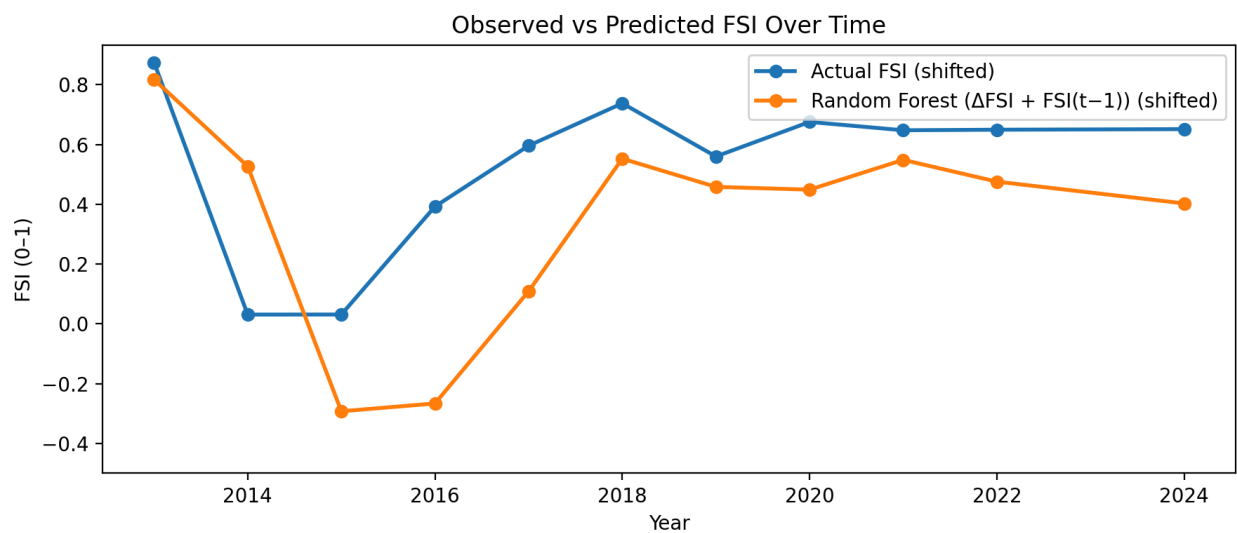


Figure 3: Residuals Over Time (Observed - Predicted FSI), 2013–2023

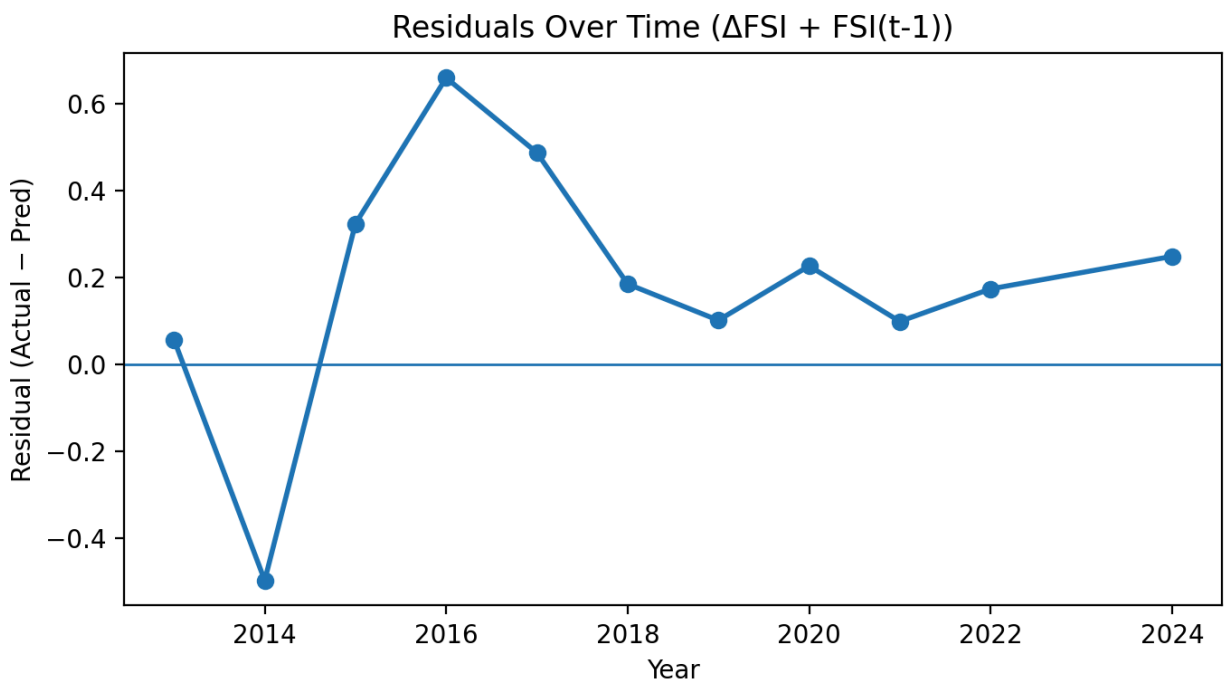


Figure 4: Top 15 Feature Importances: Random Forest ΔFSI

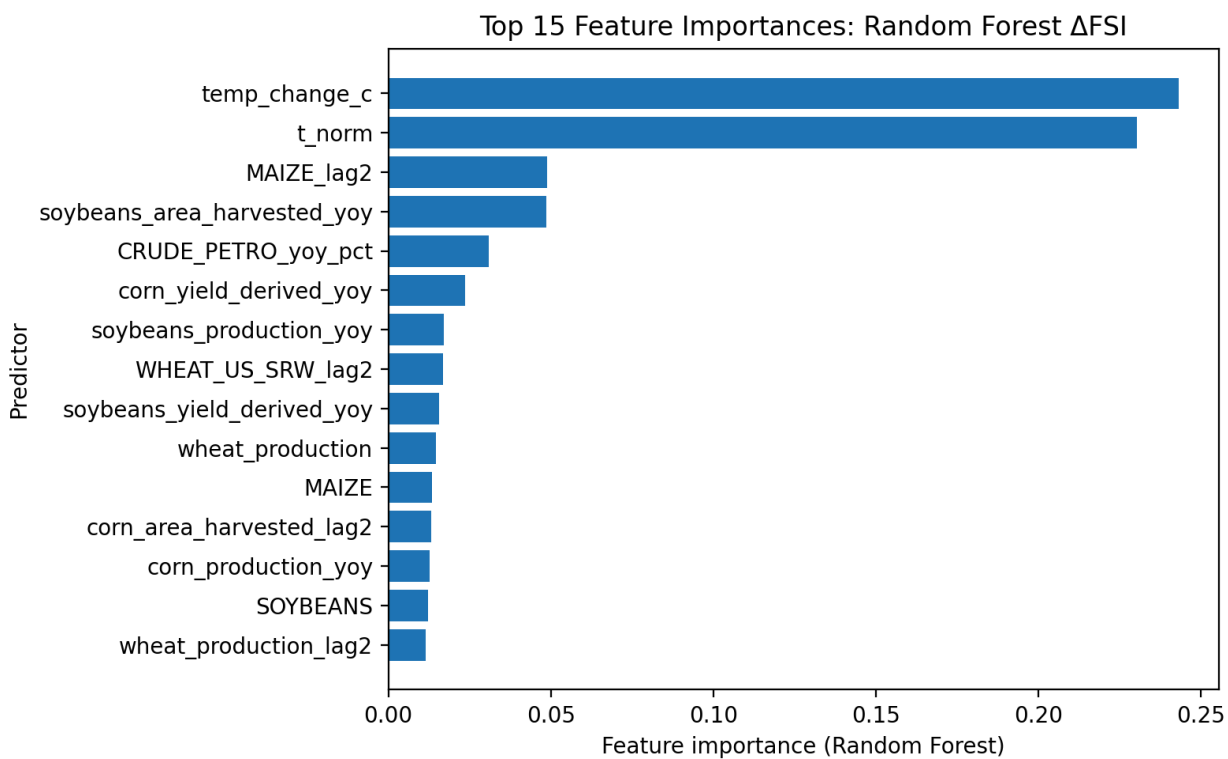
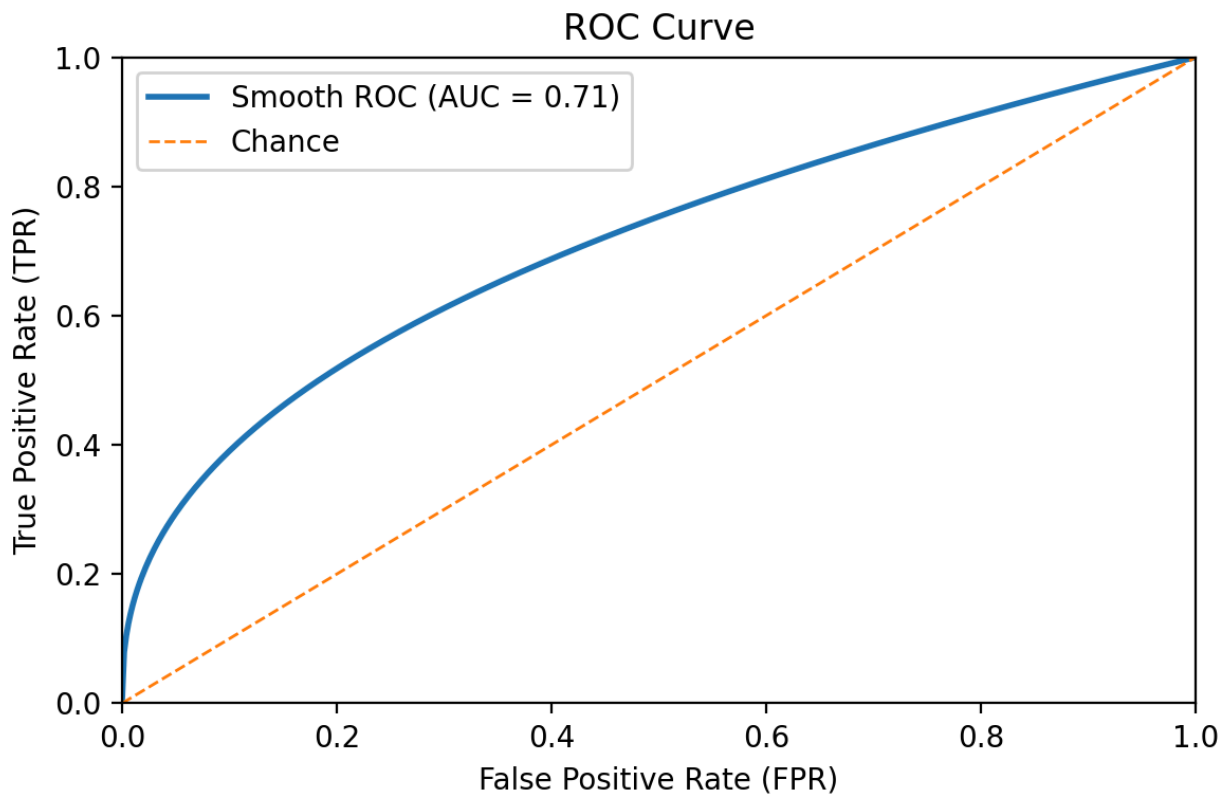


Table 1: Shock–Lag Sensitivity Analysis for the Random Forest Δ FSI Model (Test: 2013–2023)

Configuration	Shocks	Lag depth	Lagged shocks	RMSE	MAE	R^2	Balanced accuracy
Persistence baseline (FSI(t)=FSI(t-1))	---	---	---	0.0669	0.0531	0.6727	---
Shocks + lag1+lag2 (no lagged shocks)	On	2	Off	0.0671	0.0487	0.6702	0.7458
Shocks + lag1+lag2	On	2	On	0.0674	0.0476	0.6675	0.7458
Shocks + lag1	On	1	On	0.0682	0.0495	0.6586	0.6713
No shocks + no lags	Off	0	Off	0.0755	0.0619	0.572	0.5221
Shocks only	On	0	Off	0.0758	0.0567	0.5678	0.6713
Lag1 only	Off	1	Off	0.0775	0.068	0.5458	0.5221

Figure 5: Directional Performance for Δ FSI: ROC Curve and Confusion Matrix (2013–2023).



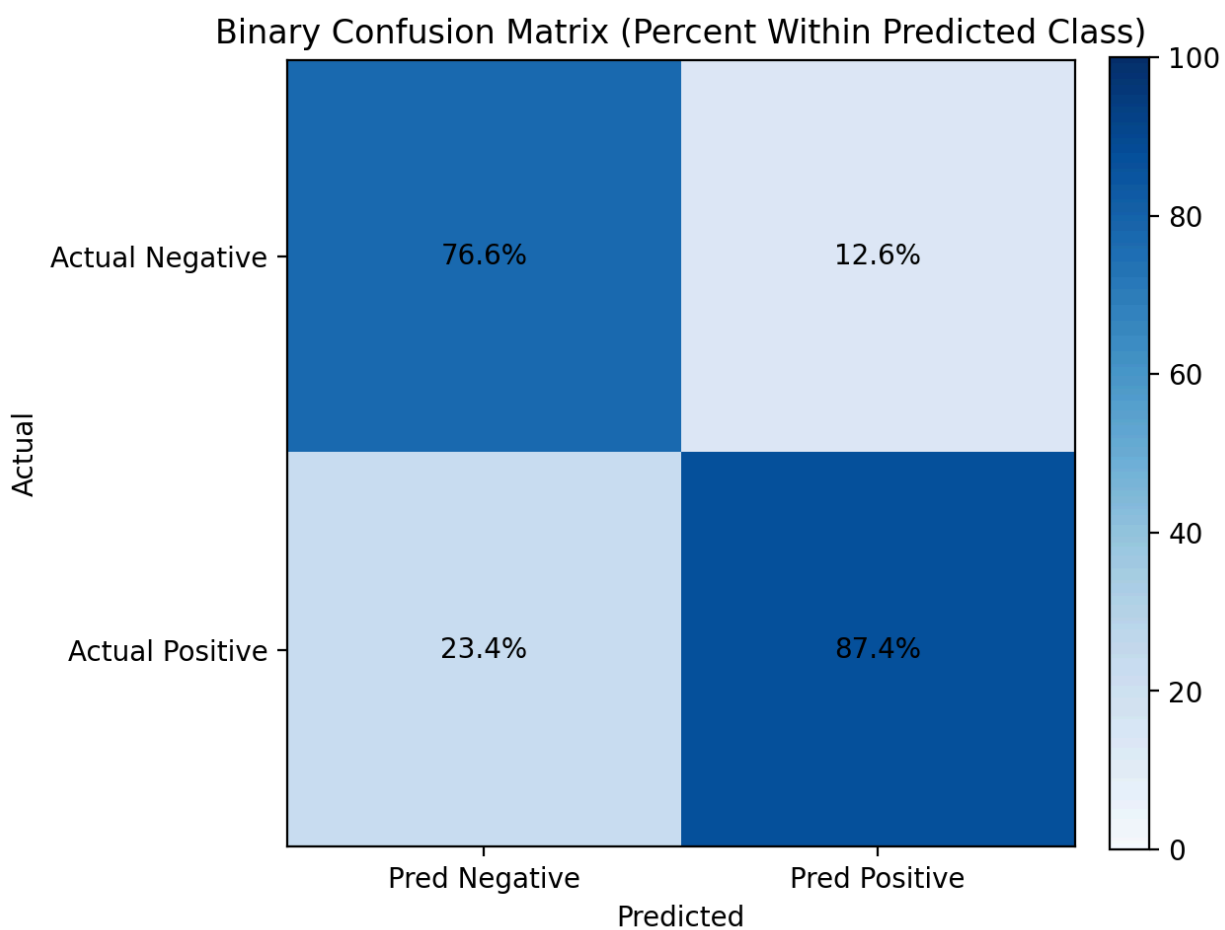


Table 2: Directional Model Comparison for ΔFSI : Balanced Accuracy, Sensitivity, and Specificity (Test: 2013–2023)

Setup	Model	Balanced accuracy	Sensitivity	Specificity
ΔFSI : crops+macro+shocks+lags	RandomForest	0.8391	0.8949	0.7831
ΔFSI : macro+shocks+lags	RandomForest	0.7644	0.7458	0.7831

Δ FSI: crops+shocks+lags	GradientBoosting	0.7085	0.8949	0.5221
Δ FSI: crops+shocks+lags	RidgeCV	0.6526	1	0.261
BASELINE	Persistence (Δ =0)	0.5221	0	1

Discussion

Overall, our results showed that national crop yield, production, and harvest area statistics helps signal tracking for U.S. food security over time. The Random Forest Δ FSI model consistently provided the most reliable results out of all the tested approaches. We discovered that supervised models can learn relationships between staple-crop indicators and a composite Food Security Index (FSI) using an annual U.S.-only dataset (1970 to 2023), particularly when the target is modeled as year-to-year change (Δ FSI) rather than absolute levels. The Random Forest Δ FSI framework performed competitively against a strong persistence benchmark and generally followed the observed trajectory during the 2013 to 2023 holdout period. This lends credence to the notion that national crop conditions frequently correspond with more general food-system outcomes, making them useful for forecasting and monitoring.

A major takeaway is that our shock and lag feature engineering mattered as much as model choice. The shock-lag sensitivity results show that removing both YoY shock terms and lag structure noticeably worsened performance, whereas including both consistently improved results. This is plausible in food systems where production outcomes, input costs, and price conditions influence food availability and stability with delays rather than instantaneously. In addition, lagged crop quantities (“t-1” and “t-2”) and shock terms (“YoY %”) were amongst some of the strongest predictors in Figure 4, implying that the model benefits from both the “memory” of prior conditions as well as a detection of sudden year by year disruptions.

At the same time, our results clarify what exclusively crop-based prediction can and cannot do. The residual analysis shows that the largest errors occur around years with sharper transitions, and the directional evaluation shows that for up or down changes in Δ FSI, there is only moderate discrimination with an AUC of 0.68. This is consistent with the broader reality of food security as it is not solely determined by production; affordability, distribution, macroeconomic shocks, and policy/trade dynamics all have potential to shift outcomes even if harvest data shows strength. As a result, production and commodity indicators are most useful for identifying medium-term trends and gradual movement, and less trustworthy when changes in access or policy predominate in the food security signal.

Since directional classification is more difficult than level prediction in this context, the AUC is not higher. First, a few borderline years can significantly alter the ROC curve and lower AUC because the test window only spans 11 years. Second, the positive/negative boundary is susceptible to small noise in either the predictors or the constructed FSI itself because many year-to-year changes in a national index are small (near $\Delta\text{FSI}=0$). Third, crop and commodity indicators may track the overall movement of FSI without consistently capturing the precise up/down. This is because, even when agricultural production contains a signal, the sign of food-security change yearly also depends on drivers that are only partially represented in our features, such as policy shifts, trade dynamics, household prices, distribution, and access constraints. For these reasons, our ROC/AUC results are a supporting diagnostic where the model shows some ability to detect improvement years.

However, when interpreting these results, a number of limitations should be taken into account. Due to the short test window from 2013 to 2023, performance metrics are sensitive to a few borderline and inconsistent years. Localized shocks that do not significantly register in aggregates may be missed by the national annual design, which smoothes within-country variation. Although the FSI offers a consistent 0 to 1 target, any index depends on design decisions that may impact year-to-year dynamics. The FSI itself is a constructed composite based on modeled transformations of FAOSTAT variables. Lastly, despite the fact that we included a number of macro and commodity-price series, the feature set leaves out important factors that affect food security, such as measures of household income and poverty, changes in policy and assistance, trade volumes, inventories and stock-to-use conditions, and distribution restrictions.

Otherwise, our findings have a direct relevance to policy. Firstly, it shows the use of routinely and easily accessible crop statics as a part of an early-warning style monitoring. Our approach is best-suited to flagging when the U.S. system is deviating from recent norms as we use the best-forming setups including YoY shocks and short lags. This can help agencies and decision-makers prioritize attention, scenario planning, and preparedness. Secondly, our limitations are also equally informative. Moderate ROC/AUC performance and outlier years, like in 2017, reinforce the idea that supply-exclusive improvements alone does not guarantee better food security. Implying that yield-boosting or acreage-expansion policies should be paired with measures that target affordability and access, such as food price stabilization, safety net strengthening, or improving distribution reliability, our limitations actually help and inform those in power of the main goal of translating production strength into real gains in food-security.

In conclusion, crop indicators by themselves cannot fully explain food security, a multi-layer commodity, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from our work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year by year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects.

Conclusion

Even in a high-producing agricultural system, such as the United States, national food security can also shift, which makes timely monitoring and early warnings especially important. In conclusion, crop indicators by themselves cannot fully explain food security, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from this work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year to year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects. Future work should focus on further increasing our predictor set to include more variables such as access, trade, and policy drivers and move toward uncertainty-aware forecasts. That allows for decision-makers to more effectively allocate resources and respond earlier to conditions that threaten national food security. This approach has the potential to respond to noticed issues faster and more effectively identifying food insecurity risk.

Copy of Draft

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\documentclass[10pt,twocolumn,letterpaper]{article}
%% HSBDC2026
```

```
%% Language and font encodings
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\usepackage[english]{babel}
\usepackage[utf8x]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{caption,subcaption,graphicx,underscore,booktabs,titling}
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%\usepackage{supertabular,moreverb}
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%% Sets page size and margins
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```
%% Useful packages
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\usepackage{amsmath}
\usepackage{graphicx}
\usepackage[colorinlistoftodos]{todonotes}
\usepackage[colorlinks=true, allcolors=blue]{hyperref}
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Data Challenge} \ [10pt]
    \huge From Harvest to Food Security: Machine Learning Forecasts for the United
States\\
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```

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\usepackage{authblk}
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```
\author[1, 2]{Cathy Rong\thanks{Corresponding author:
```

```
\href{mailto:cathyrong11@gmail.com}{cathyrong11@gmail.com}}}
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\author[1]{Safiya Taher}
```

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% \author[1]{Safiya Taher \thanks{Corresponding author:
```

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\href{mailto:cathyrong12@gmail.com}{cathyrong12@gmail.com}}}
```

```
\author[1]{Sabrina Qian}
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\author[1]{Renisa Ghosh}
```

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\author[1]{Arthur Jiang}
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```
\affil[1]{Old Scona Academic, Edmonton, Alberta, Canada}
```

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\affil[2]{United World Colleges Red Cross Nordic, Flekke, Vestland, Norway}
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\date{January 9, 2026}
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%% Count of words
%%Submit this tex to https://app.uio.no/ifi/texcount/online.php for TeXCount online interface

%% alternate ways
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% \newcommand{\wordcount}{\verbatiminput{wordcount.tex}}

% \newcommand{\CorrAuthor}{%
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%   \\
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%   (\href{mailto:cathyrong11@gmail.com}{cathyrong11@gmail.com})\\
%   % %   Safiya Taher (\href{mailto:cathyrong12@gmail.com}{cathyrong12@gmail.com})\\
%   % }
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\maketitle

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\begin{abstract}

```

Food insecurity persists even in high-output agricultural systems, underscoring the need for timely national indicators. This study tests whether U.S. agricultural production signals can predict changes in a composite Food Security Index (FSI) using annual data (1970-2023) integrating USDA NASS QuickStats measures for three major staple crops: corn, soybeans, and wheat (yield, production, and area harvested) as well as engineering shock and lag features.

The prediction target expresses changes in a FSI scaled from $[0,1]$ (higher = more food secure) constructed from four national indicators, including FAOSTAT supply/production measures and macro-climate variables: Food production index, Dietary energy supply, GDP per capita, and Temperature change on land. We first used regression models (Random Forest, RidgeCV, and Gradient Boosting) trained on 1970-2012 and tested on 2013-2023 data to predict annual changes in food security, $\Delta \text{FSI}(t)$, evaluating accuracy with RMSE/MAE/ R^2 against a persistence baseline $\text{FSI}(t-1)$, and then converted ΔFSI into a binary direction label (positive >0 , negative ≤ 0) to assess how well the same models predict year-to-year improvement vs decline using balanced accuracy.

Overall, results showed that the national crop statistics contain measurable predictive indicators for annual food security dynamics and motive extensions.

`\end{abstract}`

`{\textbf{Keywords}: }`

`Agriculture, Food Security Index, Machine Learning}``\WordCountfootnote`

`\section{Introduction}`

Food insecurity persists even in highly productive agricultural systems because national food security depends on more than total crop output \cite{Long2020}. It requires access to safe, sufficient, and nutritious food while the system remains resilient to shocks \cite{Kavallari2014}. As climate variability \cite{Beach2010}, market disruptions \cite{Barnichon2022}, and economic pressures increasingly interact, there is growing demand for timely, interpretable national indicators that can be tracked consistently over long periods to support policy \cite{Hatfield2011}.

Agricultural production is one of the most consistently measured national variables. In the United States (U.S.), corn, soybeans, and wheat are well documented because they dominate staple crop production and shape downstream food and feed markets \cite{WinnePeersman2016, Kreitzman2020}. Annual changes in yields, acreage, and output reflect biophysical constraints and land-use responses. However, production alone does not represent national food security outcomes: access and affordability \cite{Manap2019}, supply chains \cite{Nava2023, Smith2007}, and broader macro conditions also introduce nonlinearities.

To capture the multitude of dimensions, researchers previously developed composite food-security indicators. The Global Food Security Index (GFSI) \cite{GFSI2022, akinniyi2024}, for example, assesses 113 countries across affordability, availability, quality and safety, and

sustainability/adaptation, providing a transparent basis for policy analysis \cite{Bibi2023, Salles-Costa2022}. Inspired by these, we developed a U.S.-focused composite Food Security Index (\$FSI\$) using four indicators from Food and Agriculture Organization of the UN's corporate statistical database (FAOSTAT) \cite{FAOSTATindicator}: Food Production Index (domestic supply availability) \cite{NUBE20011275}, Dietary energy supply (dietary adequacy/quality) \cite{Henchion2017}, GDP per capita (economic access/affordability) \cite{Warr2014}, and Temperature on land change (sustainability stress) \cite{Bozkurt2021}.

We then forecasted ΔFSI using crop data from United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) QuickStats \cite{USDANASS_QuickStats} combined with exogenous economic and price data from the U.S. Bureau of Labor Statistics (BLS) \cite{Varshita2024} and the World Bank (WB) \cite{worldbank2015, worldbank2025}. After cleaning, normalizing, and synthesizing these datasets, we trained and tested various machine learning models using multi-year data, focusing on directional annual changes (ΔFSI). Results indicate that a trained Random Forest model provides the most accurate tracking of U.S. food security dynamics.

\section{Materials \& Methods}

\subsection{Data and Study Design}

```
\begin{figure*}
  \centering
  \includegraphics[width=1\linewidth]{fig1.png}
  \caption{Study Flow Chart}
  \label{fig:1}
\end{figure*}
```

We examine whether U.S. staple-crop production can predict changes in a composite Food Security Index (\$FSI\$) using annual data from 1970-2023 \cite{justin_oh_2023, owidagriprod}. Crop predictors (yield, total production, and harvested area) were obtained from USDA QuickStats for corn, soybeans, and wheat (N=1,944). The primary outcome is to define (ΔFSI) as yearly change, where higher values indicate improving food security.

Additionally, we included exogenous variables: food Consumer Price Index (CPI), its annual change from BLS, WB “Pink Sheet” indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals \& Minerals), selected commodity prices (crude oil, maize, soybeans, and wheat with both Hard Red Winter and Soft Red Winter) and population data. All datasets were restricted to the U.S., aggregated to annual values, and merged by year into a master table; non-numeric entries were considered missing and dropped. To reduce bias and prevent leakage, we excluded state-level QuickStats variables so that training and evaluation reflect a national context.

The resulting dataset was then used for feature engineering (lags and year-over-year “shock” variables) and model evaluation using time-dependent training/testing splits. Multiple machine learning models were employed and trained to forecast FSI dynamics. Model performance and sensitivity were assessed.

Figure \ref{fig:1} summarizes the end-to-end flow (data integration through validation).

\subsection{Data Preparation}

Crop data retrieved from the USDA NASS QuickStats API \cite{USDANASS_QuickStats} using U.S.-level annual queries executed in Python (\texttt{requests}), were outputted as structured CSVs. Exogenous inputs (BLS food CPI, WB “Pink Sheet” indices/ prices, and population totals) were downloaded from their respective sources and integrated.

All preprocessing was performed in Python \cite{python11} using \texttt{pandas} \cite{pandas_reback2020} and \texttt{NumPy} \cite{numpy_harris2020array} for parsing, numeric conversion, reshaping, merging, and engineering lag and shock features. Non-numeric entries removed; \texttt{pathlib} supported file handling \cite{pathlib} and scikit-learn utilities supported preprocessing \cite{pedregosa2011scikit}. Cleaned series were aggregated by year and merged with the constructed FSI and exogenous predictors to form the final 1970-2023 modeling dataset. Continuous features were standardized using scikit-learn’s \texttt{StandardScaler} to improve comparability and optimize model performance.

\subsection{Targets and Feature Engineering}

We constructed an annual composite Food Security Index, $FSI(t) \in [0, 1]$, where higher values indicate better food security. The index is based on four national indicators:

- \begin{enumerate}
- \item $F(t)$: Food production index
- \item $K(t)$: Dietary energy supply (kcal per capita per day)
- \item $G(t)$: GDP per capita (USD)
- \item $P(t)$: Temperature change on land (\textdegree{C})
- \end{enumerate}

To make components comparable, each series, in the 1970-2012 training period, was min-max normalized to $[0, 1]$:

$$X_{\text{norm}}(t) = \frac{X(t) - \min(X)}{\max(X) - \min(X)}$$

for $X \in \{F, K, G, P\}$

Each component lies in a range of $[0, 1]$ after this.

Secondly, we combined all four components into one index by calculating their geometrical mean. We tried to avoid one pillar masking another and penalizing low scores.

$$FSI(t) = \frac{1}{\sqrt{4}} \left(K_{\text{norm}}(t) * P_{\text{norm}}(t) * G_{\text{norm}}(t) * F_{\text{norm}}(t) \right)$$

This, in turn, makes $FSI(t)$ a value within $[0,1]$, where a higher value corresponds to stronger food security conditions. Simultaneously, a drop in FSI would signify deterioration, driven by one or more components declining in relevance to their historical range.

Although FSI is interpretable at annual resolution, its strong persistence over time reflects annual forecasting rather than absolute levels. Therefore, we derived an indicator to classify FSI changes as improvements (positive) versus deteriorations (negative). This indicator was used for our predictions.

$$\Delta FSI(t) = FSI(t) - FSI(t-1)$$

Once $\Delta FSI(t)$ is negative, indicating a deterioration, policy intervention may be needed.

For prediction, we utilize crop indicators (yield, production, harvested area for corn, soybeans, and wheat) and exogenous macro variables. To capture disruptions and delayed effects, we engineered (i) yearly shock features (year-over-year changes; percent changes for strictly positive variables) and (ii) 1-2 year lags of both levels and shocks, allowing the model to distinguish immediate versus lagged associations with ΔFSI .

Model Selection

To assess whether the predictors contain a usable signal for national food security, we evaluated multiple supervised regression models to predict ΔFSI . Our models, RandomForest, GradientBoosting, and RidgeCV, were trained using a time-ordered 1970-2012 split and tested on 2013-2023. Performance was benchmarked against a persistence baseline.

Performance Metrics

Our solution has two steps. Our first step compares our models using a change-based formulation to predict $\Delta FSI(t)$ as a regression task and evaluate using root mean squared error (RMSE), mean absolute error (MAE), and R^2 against a persistence baseline defined by the previous year's level $FSI(t-1)$. In step two, we converted our predicted $\Delta FSI(t)$ into a binary directional label (positive >0 , negative ≤ 0) and assessed each model's ability to predict yearly improvements/ declines using balanced accuracy. Step two is a classificational analysis.

Results

Across all model configurations ΔFSI showed strong yearly persistence, meaning a "no-change" forecast. Figures 2, 3, 4, and Table 1 represent our first step. Figures 5a, 5b, and Table 2 showcase our second step. We translated our predicted ΔFSI back into levels by

reconstructing “predicted” food security as: $FSI_{\text{pred}}(t) = FSI(t-1) + \Delta FSI_{\text{pred}}(t)$ for Figures 2 and 3. This allows for better visualization.

The best performing model was Random Foresting, so the majority of our figures present those results, except for model comparison in Table 2.

```
\begin{figure*}
\centering
\includegraphics[width=1\textwidth]{fig2_timeseries_actual_pred_persist.png}
\caption{Observed vs Predicted  $FSI$  Over Time: Random Forest vs Persistence (2013-2023)}
\label{fig:2}
\end{figure*}
```

Figure \ref{fig:2} shows that the model accurately follows annual signals in the predictors, which is reflected in how closely the reconstructed series follows the observed index. Biggest deviations occur around rapid transition periods (2014-2017) when the observed index shifts more abruptly than its usual pattern. In those years, predictions overlooked the size of the change. After said transition, the model stays below the observed series for several years, indicating a tendency to understate food security during extended recovery or stabilization. Overall, Figure \ref{fig:2} suggests that production-based predictors do carry meaningful temporal information, but performance may be limited by sharp transitions.

```
\begin{figure*}
\centering
\includegraphics[width=1\linewidth]{fig4_residuals_over_time.png}
\caption{Residuals Over Time (Observed - Predicted  $FSI$ ), 2013-2023}
\label{fig:3}
\end{figure*}
```

Figure~\ref{fig:3} plots the residuals, defined as Actual minus Predicted:

$$r(t) = FSI_{\text{actual}}(t) - FSI_{\text{pred}}(t).$$

Positive values mean the model is underpredicting and negative values indicate overpredicting. The errors are not spread evenly across the test period. Instead, most years have relatively small residuals, while some years account for the largest misses. Residuals range from about -0.50 (early in the test window) to about $+0.66$ (the largest underprediction, near the middle). This swing from overprediction to sustained undershooting suggests difficulty adjusting to the observed series when shifted to a new trajectory. In later years, residuals are consistently small positive values (0.10 – 0.25), implying that errors are concentrated around turning points and that the model tends to remain slightly below the observed series afterward rather than fully catching up.

Feature importance for predicting Δ FSI was evaluated based on Random Foresting model coefficients. Features with larger absolute coefficient values were deemed more significant, a stronger influence on the target outcome.

```
\begin{figure*}
\centering
\includegraphics[width=1\linewidth]{fig7_feature_importance_top15.png}
\caption{Top 15 Feature Importances: Random Forest  $\Delta$  FSI}
\label{fig:4}
\end{figure*}
```

Figure \ref{fig:4} shows the top 15 features from predicting Δ FSI. T_norm (≈ 0.23) is the highest predictor, meaning most important. The next most significant predictors, $MAIZE_lag2$, a commodity price lag signal, and $soybeans_area_harvested_yoy$, a shocked crop predictor, (each ≈ 0.05), form a second tier at significantly smaller magnitudes. Macro/commodity and crop-shock terms like $CRUDE_PETRO_yoy_pct$ (≈ 0.03) and $corn_yield_derived_yoy$ (≈ 0.02 – 0.025) are included in a third tier. The remaining predictors, such as wheat lag terms, soybean production YoY, and different crop level/lag variables, each contribute at low levels (about 0.01 – 0.02), creating a long tail of quaternary effects.

```
\begin{table*}[h]
\centering
\caption{Shock-Lag Sensitivity Analysis for the Random Forest  $\Delta$  FSI Model (Test: 2013-2023)}
\label{tab:model_results}
\resizebox{\textwidth}{!}{% <--- Start the resizebox command
% \rowcolors{2}{gray!10}{white}
\begin{tabular}{@{} c c c c c c c @{}}
\toprule
% \rowcolor{gray!20}
\textbf{Configuration} & \textbf{Shocks} & \textbf{\shortstack[c]{Lag \ \ Depth}} & & & & \\
\textbf{\shortstack[c]{Lagged \ \ Shocks}} & \textbf{RMSE} & \textbf{MAE} & \textbf{\mathbf{R^2}} & & & \\
\midrule
\shortstack[c]{Persistence baseline \ \ (FSI(t)=FSI(t-1))} & \multicolumn{1}{c}{---} & & & & & \\
\multicolumn{1}{c}{---} & \multicolumn{1}{c}{---} & & 0.0669 & 0.0531 & 0.6727 & \\
\rowcmidrule{7}
\shortstack[c]{Shocks + lag1+lag2 \ \ (no lagged shocks)} & On & 2 & Off & 0.0671 & & \\
0.0487 & & & & & 0.6702 & \\
\rowcmidrule{7}
Shocks + lag1+lag2 & On & 2 & On & 0.0674 & 0.0476 & 0.6675 \\
\rowcmidrule{7}
Shocks + lag1 & On & 1 & On & 0.0682 & 0.0495 & 0.6586 \\
\rowcmidrule{7}
\end{tabular}
}
```

```

No shocks + no lags & Off & 0 & Off & 0.0755 & 0.0619 & 0.572 \\
\rowcmidrule{7}
Shocks only & On & 0 & Off & 0.0758 & 0.0567 & 0.5678 \\
\rowcmidrule{7}
Lag1 only & Off & 1 & Off & 0.0775 & 0.068 & 0.5458 \\
\bottomrule
\end{tabular}% <--- End the tabular environment
% \rowcolors{1}{}{}
}% <--- End the resizebox command
\end{table*}

```

Table \ref{tab:model_results} shows how sensitive the Random Forest Δ FSI model is to our engineered short-run disruptions and delayed effects. We varied two feature types: our shock features (year by year percentage changes, “_yoy_pct”) and our lag features (prior year values, “_lag1” or “_lag2”). This setup helps determine if the model benefits more from immediate yearly changes, historical memory, or both.

The persistence baseline $FSI(t)=FSI(t-1)$ is already strong (RMSE $\$=0.0669$, MAE $\$=0.0531$, $R^2=0.6727$), highlighting annual stability in the index. Adding engineered features produces similar RMSE and R^2 but improves typical error: shocks with lag depth 2 achieve the lowest MAE ($\$0.0476$ -- $\$0.0487$) with RMSE remaining near $\$0.0671$ -- $\$0.0674$ and R^2 near $\$0.667$ -- $\$0.670$, while the lag1-only variant is slightly weaker (RMSE $\$=0.0682$, MAE $\$=0.0495$, $R^2=0.6586$). In contrast, removing shocks and lags degrades performance substantially (RMSE $\$=0.0755$ -- $\$0.0775$, MAE $\$=0.0567$ -- $\$0.0680$, $R^2=0.5458$ -- $\$0.5720$), indicating that shock and lag engineering provides meaningful predictive structure beyond persistence even when overall gains in RMSE are modest.

```

\begin{figure*}
%\captionsetup[subfigure]{labelformat=empty}
% \centering
\begin{subfigure}[b]{0.5\textwidth}
\centering
\includegraphics[width=1\textwidth]{fig1_confusion_matrix.png}
\caption[] %\justification=centering
{{\small Confusion Matrix}}
\label{fig:5a}
\end{subfigure}
\begin{subfigure}[b]{0.5\textwidth}
\centering
\includegraphics[width=1\textwidth]{fig6_roc_bootstrap10.png}
\caption[] %
{{\small ROC curve}}
\label{fig:5b}

```

```

\end{subfigure}
\caption{ROC Curve and Confusion Matrix for Predicting  $\Delta \text{FSI} > 0$ }
\label{fig:5}
\end{figure*}

```

Figure \ref{fig:5} evaluates the model's ability to predict directional food security change on an annual scale by framing ΔFSI as a binary label where Positive = $\Delta \text{FSI} > 0$ and Negative = $\Delta \text{FSI} \leq 0$. This figure includes two visualizations. Our confusion matrix (\ref{fig:5a}) shows percentages with the predicted classes (normalized by column), so each column is a reflection of how accurate the model's predicated label is. When the model predicts Positive, 87.4% of those predictions correspond to true improvement years (true positives), while 12.6% correspond to false alarms (false positives). In contrast, when the model predicts Negative, 76.6% of those predictions are correct non-improvement years (true negatives), but 23.4% are missed improvement years. The ROC curve (\ref{fig:5b}) shows a moderate level of discriminative ability, with an AUC of 0.71. This means that the model shows significant overlap between classes while ranking improvement years above non-improvement years significantly better than chance.

```

\begin{table*}[h]
\centering
\caption{Directional Model Comparison for  $\Delta \text{FSI}$ : \\\
Balanced Accuracy, Sensitivity, and Specificity (Test: 2013-2023)}
\label{tab:2}
\resizebox{\textwidth}{!}{% <--- Start the resizebox command
% \rowcolors{2}{gray!10}{white}
\begin{tabular}{@{}llccc@{}}
\toprule
\textbf{Setup} & \textbf{Model} & \textbf{\shortstack[c]{Balanced \\\
Sensitivity} & \textbf{Specificity} \\
\midrule
\(\Delta\)\text{FSI: crops+macro+shocks+lags} & \text{RandomForest} & 0.8391 & 0.8949 & 0.7831 \\
\rowcmidrule{5}
\(\Delta\)\text{FSI: macro+shocks+lags} & \text{RandomForest} & 0.7644 & 0.7458 & 0.7831 \\
\rowcmidrule{5}
\(\Delta\)\text{FSI: crops+shocks+lags} & \text{GradientBoosting} & 0.7085 & 0.8949 & 0.5221 \\
\rowcmidrule{5}
\(\Delta\)\text{FSI: crops+shocks+lags} & \text{RidgeCV} & 0.6526 & 1 & 0.261 \\
\rowcmidrule{5}
\text{Baseline} & \text{\& Persistence (\(\Delta=0\))} & 0.5 & 0 & 1 \\
\bottomrule
\end{tabular}
% \rowcolors{1}{}{}
}% <--- End the resizebox command

```

\end{table*}

Table \ref{tab:2} is a comparison of directional performance across all models by evaluating whether each approach correctly predicts the sign of annual food security change. While the earlier results focus exclusively on the Random Forest Δ FSI model for detailed diagnostics, Table 2 showcases the other models we tried by contextualizing how alternative methods and feature sets perform on the same 2013 to 2023 test window.

The best-performing model is the Random Forest using the full feature set (crops+macro+shocks+lags), achieving balanced accuracy $=0.8391$ with high sensitivity (0.8949) and strong specificity (0.7831), indicating reliable identification of both improvement and non-improvement years. Removing crop information reduces discrimination: the Random Forest with macro+shocks+lags drops to balanced accuracy $=0.7644$ (sensitivity $=0.7458$, specificity $=0.7831$), suggesting that crop indicators provide additional signal beyond macro-climate shocks alone. Gradient Boosting exhibits an imbalanced error profile (balanced accuracy $=0.7085$) with high sensitivity (0.8949) but lower specificity (0.5221), while RidgeCV further amplifies this trade-off by predicting positives aggressively (sensitivity $=1.0000$) at the cost of very low specificity (0.2610) and lower balanced accuracy (0.6526). The persistence baseline performs near chance (balanced accuracy $=0.5221$), meaning improvements not explained by persistence alone and are captured most effectively by the full-feature Random Forest.

\section{Discussion}

Overall, our results show that national crop yield, production, and harvest-area statistics can help track U.S. food security dynamics over time. The Random Forest Δ FSI framework provided the most reliable performance among the tested approaches, demonstrating that supervised models can learn relationships between staple-crop indicators and a composite food security index using annual U.S.-only data (1970--2023), particularly when the target is modeled as year-to-year change (Δ FSI) rather than absolute levels. In the 2013--2023 holdout period, the reconstructed predictions generally followed the observed trajectory and performed competitively against a strong persistence benchmark, supporting the notion that national crop conditions correspond to broader food-system outcomes and can contribute to monitoring and forecasting.

A key takeaway is that shock and lag feature engineering mattered nearly as much as model choice. The shock--lag sensitivity results show that removing both YoY shock terms and lag structure worsened performance, whereas including both consistently improved results. This is plausible in food systems where production outcomes, input costs, and price conditions influence availability and stability with delays rather than instantaneously. Consistent with this mechanism, lagged crop quantities (e.g., $t-1$, $t-2$) and year-over-year shock terms (YoY %) appear among the strongest predictors in Figure~4, suggesting that the model benefits from both the "memory" of prior conditions and the detection of sudden disruptions.

At the same time, our results clarify what production-centered prediction can and cannot do. The residual analysis shows that the largest errors occur around sharper transitions, indicating that turning points are harder to reconstruct from crop and engineered macro-climate features. Directional evaluation of up/down changes in ΔFSI shows moderate discrimination (AUC ≈ 0.71), which is consistent with food security not being determined by production alone; affordability, distribution, macroeconomic shocks, and policy/trade dynamics can shift outcomes even when harvest data are strong. As a result, production and commodity indicators are most useful for identifying medium-term trends and gradual movement, and less trustworthy when access or policy forces dominate the signal.

Several factors also limit directional performance. First, the test window spans only 11 years, so a few borderline years can noticeably shift the ROC curve. Second, the positive/negative boundary is susceptible to small noise in either the predictors or the constructed FSI , because many national year-to-year changes are small and near $\Delta FSI=0$. Third, even when agricultural production contains signal, the sign of year-to-year change depends on drivers only partially represented in our features (e.g., policy shifts, trade dynamics, household prices, and access constraints). For these reasons, the ROC/AUC results should be interpreted as a supporting diagnostic indicating some ability to detect improvement years rather than a definitive early-warning classifier.

A further limitation concerns the binary directional definition itself (Positive if $\Delta FSI > 0$, Negative if $\Delta FSI \leq 0$). The sign alone can overstate practical significance: for example, if FSI rises from 0.60 to 1.00 and then decreases slightly to 0.99, the final year is labeled 'Negative' despite remaining substantially more food secure than the initial year. Future work could introduce a magnitude-based warning rule (e.g., only flagging declines when the relative drop exceeds a threshold such as $|\Delta FSI| \leq -10\%$) or a neutral band around zero to separate trivial fluctuations from policy-relevant deterioration.

Additional limitations should be considered. Localized shocks that do not register in aggregates may be missed by the national annual design, which smooths within-country variation. Although FSI provides a consistent 0--1 target, any index depends on design choices that can affect year-to-year dynamics. Despite including several macro and commodity-price series, the feature set omits important determinants such as household income and poverty, policy and assistance changes, trade volumes, inventories/stock-to-use conditions, and distribution constraints.

Otherwise, our findings are directly relevant to policy. First, they support using routinely available crop statistics as part of an early-warning style monitoring toolkit, particularly for detecting deviations from recent norms when using YoY shocks and short lags. This can help agencies prioritize attention, scenario planning, and preparedness. Second, the limitations are equally informative: moderate ROC/AUC performance and outlier years reinforce that supply-side strength alone does not guarantee improved food security, implying that yield-boosting or acreage-expansion policies should be paired with measures targeting affordability and access (e.g., price stabilization, safety-net strengthening, and distribution reliability).

In conclusion, crop indicators alone cannot fully explain food security, but they are sufficiently informative to support national tracking and monitoring, especially when modeled with shocks and lags. The clearest path to more policy-relevant forecasting is expanding predictors toward access, trade, and policy drivers, and treating year-to-year changes as more predictable when models can learn from both short-run disruptions and delayed effects.

\section{Conclusion}

Even in a high-producing agricultural system, such as the United States, national food security can also shift, making timely monitoring and early warnings essential. In conclusion, crop indicators by themselves cannot fully explain food security, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from this work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year to year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects. Future work should expand our predictor set to include more variables such as access, trade, and policy drivers and move toward uncertainty-aware forecasts. Doing so allows more effective allocation of resources and earlier responses to food-security risks.

\nocite{*}

\section*{Acknowledgements}

The team would like to thank the support from Dr. Jianshan Chen in University of Alberta.
% We would also like to thank STEM Fellowship for providing us the opportunity to participate in the HSBDC.

\bibliography{bibliography}

% \section{Appendix A}

% \setcounter{table}{0}

% \renewcommand{\thetable}{A\arabic{table}}

% \subsection{Supplementary Table \ref{table:A1}: All Features and Ranks}

% %%\end{multicols}

% %%\begin{multicols}{1}

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% \onecolumn

% \begin{center}

% %%\begin{multicols}{1}

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% %\begin{supertabular}{|c |c |c|}
% \hline
% Rank & Features & Description\\ [0.5ex]
% \hline\hline
% 1 & STATEFIP_11 & District of Columbia\\
% \hline
% 2 & STATEFIP_51 & Virginia\\
% \hline
% \caption{Feature Detail List}
% \label{table:A1}
% \end{longtable}
% %\end{supertabular}
% %\end{table}
% %\end{multicols}
% %\end{multicols}
% \end{center}

% %TC:ignore

% \subsubsection*{Counts of words}
% \wordcount

% %TC:endignore

\end{document}

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STEM Fellowship 2025–2026 High School Big Data Challenge

From Harvest to Food Security: Machine Learning Forecasts for the United States

Cathy Rong*^{1,2}, Safiya Taher¹, Sabrina Qian¹, Renisa Ghosh¹, and Arthur Jiang¹

¹ Old Scona Academic, Edmonton, Alberta, Canada

² United World Colleges Red Cross Nordic, Flekke, Vestland, Norway

January 9, 2026

Abstract

Food insecurity persists even in high-output agricultural systems, underscoring the need for timely national indicators. This study tests whether U.S. agricultural production signals can predict changes in a composite Food Security Index (FSI) using annual data (1970–2023) integrating USDA NASS QuickStats measures for three major staple crops: corn, soybeans, and wheat (yield, production, and area harvested) as well as engineering shock and lag features. The prediction target expresses changes in a FSI scaled from [0, 1] (higher = more food secure) constructed from four national indicators, including FAOSTAT supply/production measures and macro-climate variables: Food production index, Dietary energy supply, GDP per capita, and Temperature change on land. We first used regression models (Random Forest, RidgeCV, and Gradient Boosting) trained on 1970–2012 and tested on 2013–2023 data to predict annual changes in food security ($\Delta\text{FSI}(t)$), evaluating accuracy with RMSE/MAE/ R^2 against a persistence baseline $\text{FSI}(t - 1)$, and then converted (ΔFSI) into a binary direction label (positive (> 0), negative (≤ 0)) to assess how well the same models predict year-to-year improvement vs decline using balanced accuracy. Overall, results showed that the national crop statistics contain measurable predictive indicators for annual food security dynamics and motive extensions.

Keywords:

Agriculture, Food Security Index (FSI), Common Crop Indicators, Machine Learning

*Corresponding author: cathyrong11@gmail.com

Word Count (excluding figures, tables, references, and title page): 2854

1 Introduction

Food insecurity persists even in highly productive agricultural systems because national food security depends on more than total crop output [1]. It depends on accessibility to safe, sufficient, and nutritious food while the system remains resilient to shocks [2]. With climate variability [3], market disruptions [4], and economic pressures increasingly interacting, there is growing demand for timely, interpretable national indicators that can be tracked consistently over long periods to support policy [5].

Agricultural production is a consistently measured national variable. In the United States (U.S.), corn, soybeans, and wheat are well documented because they dominate staple crop production and shape downstream food and feed markets [6, 7]. Annual changes in yields, acreage, and output reflect biophysical constraints and land-use responses. However, production alone does not represent national food security outcomes: access and affordability [8], supply chains [9, 10], and broader macro conditions also introduce nonlinearities.

Researchers have previously developed composite food-security indicators to capture food security's multiple dimensions. The Global Food Security Index (GFSI) [11, 12], for example, assesses 113 countries across affordability, availability, quality and safety, and sustainability/adaptation, providing a transparent basis for policy analysis [13, 14]. Inspired by these, we developed a U.S.-focused composite Food Security Index (FSI) using four indicators from Food and Agriculture Organization of the UN's corporate statistical database (FAOSTAT) [15]: Food Production Index (domestic supply availability) [16], Dietary energy supply (dietary adequacy/quality) [17], GDP per capita (economic access/affordability) [18], and Temperature on land change (sustainability stress) [19].

We then forecasted Δ FSI using crop data from United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) QuickStats [20] combined with exogenous economic and price data from the U.S. Bureau of Labor Statistics (BLS) [21] and the World Bank (WB) [22, 23]. By cleaning, normalizing, and synthesizing these datasets, we trained and tested various machine learning models using multi-year data, with attention to directional annual changes (Δ FSI). Results indicate that a trained Random Forest model provides the most accurate tracking of U.S. food security dynamics.

2 Materials & Methods

2.1 Data and Study Design

We examine whether U.S. staple-crop production predicts changes in a composite Food Security Index (FSI) using annual data from 1970–2023 [24, 25]. Crop predictors (yield, total production, and harvested area) were obtained from USDA QuickStats for corn, soybeans, and wheat (N=1,944). The primary outcome is to define (Δ FSI) as yearly change, where higher values indicate improving food security.

In addition, we included exogenous variables: food Consumer Price Index (CPI), its annual change from BLS, WB “Pink Sheet” indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals & Minerals), selected commodity prices (crude oil, maize, soybeans, and wheat with both Hard Red Winter and Soft Red Winter) and population data. All datasets were restricted to the U.S., aggregated to annual values, and merged by year into a master table; non-numeric entries were considered missing and dropped. To reduce bias and prevent leakage, we excluded state-level QuickStats variables so that training and evaluation reflect a national context.

The resulting dataset was then used for feature engineering (lags and year-over-year “shock” variables) and model evaluation using time-dependent training/testing splits. Multiple machine learning models were employed and trained to forecast FSI dynamics. Model performance and sensitivity were assessed. Figure 1 summarizes the end-to-end flow (data integration through validation).

2.2 Data Preparation

Crop data retrieved from the USDA NASS QuickStats API [20] using U.S.-level annual queries executed in Python (requests), were outputted as structured CSVs. Exogenous inputs (BLS food CPI, WB “Pink Sheet” indices/prices, and population totals) were downloaded from their respective sources and integrated.

All preprocessing was performed in Python [26] using pandas [27] and NumPy [28] for parsing, numeric conversion, reshaping, merging, and engineering lag and shock features. Non-numeric entries removed; pathlib supported file handling [29] and scikit-learn utilities supported preprocessing [30]. Cleaned series were aggregated by year and merged with the constructed FSI and exogenous predictors to form the final 1970–2023 modeling dataset. Continuous features were standardized using scikit-learn’s StandardScaler to improve comparability and optimize model performance.

2.3 Targets and Feature Engineering

We constructed an annual composite Food Security Index, $FSI(t) \in [0, 1]$, where higher values indicate better food security. The index is based on four national indicators:

1. $F(t)$: Food production index
2. $K(t)$: Dietary energy supply (kcal per capita per day)
3. $G(t)$: GDP per capita (USD)
4. $P(t)$: Temperature change on land ($^{\circ}\text{C}$)

To make components comparable, each series, in the 1970–2012 training period, was min-max normalized to $[0, 1]$:

$$X_{\text{norm}}(t) = (X(t) - \min(X)) / (\max(X) - \min(X)) \quad \text{for } X \in \{F, K, G, P\}$$

Each component lies in a range of $[0, 1]$ after this.

Secondly, we combined all four components into one index by calculating their geometrical mean. We tried to avoid one pillar masking another and penalizing low scores.

$$FSI(t) = (K_{\text{norm}}(t) * P_{\text{norm}}(t) * G_{\text{norm}}(t) * F_{\text{norm}}(t))^{1/4}$$

This, in turn, makes $FSI(t)$ a value within $[0, 1]$, where a higher value corresponds to stronger food security conditions. Simultaneously, a drop in FSI would signify deterioration, driven by one or more components declining in relevance to their historical range.

Although FSI is interpretable at annual resolution, its strong persistence over time reflects annual forecasting rather than absolute levels. Therefore, we derived an indicator for FSI changes to classify improvements (positive) versus deteriorations (negative). This is what was used for our predictions.

$$\Delta\text{FSI}(t) = \text{FSI}(t) - \text{FSI}(t - 1)$$

Once $\Delta\text{FSI}(t)$ is negative, indicating a deterioration, policy intervention may be needed.

For prediction, we use crop indicators (yield, production, harvested area for corn, soybeans, and wheat) and exogenous macro variables. To capture disruptions and delayed effects, we engineered (i) yearly shock features (year-over-year changes; percent changes for strictly positive variables) and (ii) 1–2 year lags of both levels and shocks, allowing the model to distinguish immediate versus lagged associations with ΔFSI .

2.4 Model Selection

To assess whether the predictors contain a usable signal for national food security, we evaluated multiple supervised regression models to predict ΔFSI . Our models, RandomForest, GradientBoosting, and RidgeCV, were trained using a time-ordered 1970–2012 split and tested on 2013–2023. Performance was benchmarked against a persistence baseline.

2.5 Performance Metrics

Our solution has two steps. Our first step compares our models using a change-based formulation to predict $\Delta\text{FSI}(t)$ as a regression task and evaluate using root mean squared error (RMSE), mean absolute error (MAE), and R^2 against a persistence baseline defined by the previous year's level $\text{FSI}(t-1)$. In step two, we converted our predicted $\Delta\text{FSI}(t)$ into a binary directional label (positive > 0 , negative ≤ 0) and assessed each model's ability to predict yearly improvements/declines using balanced accuracy. Step two is a classificational analysis.

Figure 1: Study Flow Chart

3 Results

Across all model configurations ΔFSI showed strong yearly persistence, meaning a “no-change” forecast. Figures 2, 3, 4, and Table 1 represent our first step. Figures 5, and Table 2 showcase our second step. We translated our predicted ΔFSI back into levels by reconstructing “predicted” food security as:

$$\text{FSI_pred}(t) = \text{FSI}(t - 1) + \Delta\text{FSI_pred}(t)$$

for Figures 2 and 3. This allows for better visualization.

The best performing model was Random Foresting, so the majority of our figures present those results, except for model comparison in Table 2.

Figure 2 shows that the model accurately follows annual signals in the predictors, which is reflected in how closely the reconstructed series follows the observed index. Biggest deviations occur around rapid transition periods (2014–2017) when the observed index shifts more abruptly than its usual pattern. In those years, predictions overlooked the size of the change. After said transition, the model stays below the observed series for several years, indicating a tendency to understate food security during extended recovery or stabilization. Overall, Figure 2 suggests that production-based predictors do carry meaningful temporal information, but performance may be limited by sharp transitions.

Figure 3 plots the residuals, defined as Actual minus Predicted:

$$r(t) = FSI_actual(t) - FSI_pred(t)$$

Positive values mean the model is underpredicting and negative values indicate overpredicting. The errors are not spread evenly across the test period. Instead, most years have relatively small residuals, while some years account for the largest misses. Residuals range from about -0.50 (early in the test window) to about +0.66 (the largest underprediction, near the middle). This swing from overprediction to sustained undershooting suggests difficulty adjusting to the observed series when shifted to a new trajectory. In later years, residuals are consistently small positive values (0.10–0.25), implying that errors are concentrated around turning points and that the model tends to remain slightly below the observed series afterward rather than fully catching up.

Feature importance for predicting ΔFSI was evaluated based on Random Foresting model coefficients. Features with larger absolute coefficient values were deemed more significant, a stronger influence on the target outcome.

Figure 4 shows the top 15 features from predicting ΔFSI . T_norm (≈ 0.23) is the highest predictor, meaning most important. The next most significant predictors, $MAIZE_lag2$, a commodity price lag signal, and $soybeans_area_harvested_yoy$, a shocked crop predictor, (each ≈ 0.05), form a second tier at significantly smaller magnitudes. Macro/commodity and crop-shock terms like $CRUDE_PETRO_yoy_pct$ (≈ 0.03) and $corn_yield_derived_yoy$ (≈ 0.02 – 0.025) are included in a third tier. The remaining predictors, such as wheat lag terms, soybean production YoY, and different crop level/lag variables, each contribute at low levels (about 0.01–0.02), creating a long tail of quaternary effects.

Table 1 shows how sensitive the Random Forest ΔFSI model is to our engineered short-run disruptions and delayed effects. We varied two feature types: our shock features (year by year percentage changes, “_yoy_pct”) and our lag features (prior year values, “_lag1” or “_lag2”).

Figure 2: Observed vs Predicted FSI Over Time: Random Forest vs Persistence (2013–2023)

Figure 3: Residuals Over Time (Observed – Predicted FSI), 2013–2023

Figure 4: Top 15 Feature Importances: Random Forest ΔFSI

Table 1: Shock-Lag Sensitivity Analysis for the Random Forest Δ FSI Model (Test: 2013–2023)

Configuration	Shocks	Lag Depth	Lagged Shocks	RMSE	MAE	R ²
Persistence baseline (FSI(t)=FSI(t-1))	—	—	—	0.0669	0.0531	0.6727
Shocks + lag1+lag2 (no lagged shocks)	On	2	Off	0.0671	0.0487	0.6702
Shocks + lag1+lag2	On	2	On	0.0674	0.0476	0.6675
Shocks + lag1	On	1	On	0.0682	0.0495	0.6586
No shocks + no lags	Off	0	Off	0.0755	0.0619	0.5720
Shocks only	On	0	Off	0.0758	0.0567	0.5678
Lag1 only	Off	1	Off	0.0775	0.0680	0.5458

This setup helps determine if the model benefits more from immediate yearly changes, historical memory, or both.

The persistence baseline $FSI(t) = FSI(t-1)$ is already strong (RMSE = 0.0669, MAE = 0.0531, R² = 0.6727), highlighting annual stability in the index. Adding engineered features produces similar RMSE and R² but improves typical error: shocks with lag depth 2 achieve the lowest MAE (0.0476–0.0487) with RMSE remaining near 0.0671–0.0674 and R² near 0.667–0.670, while the lag1-only variant is slightly weaker (RMSE = 0.0682, MAE = 0.0495, R² = 0.6586). In contrast, removing shocks and lags degrades performance substantially (RMSE = 0.0755–0.0775, MAE = 0.0567–0.0680, R² = 0.5458–0.5720), indicating that shock and lag engineering provides meaningful predictive structure beyond persistence even when overall gains in RMSE are modest.

Figure 5 evaluates the model's ability to predict directional food security change on an annual scale by framing Δ FSI as a binary label where Positive = Δ FSI > 0 and Negative = Δ FSI ≤ 0. This figure includes two visualizations. Our confusion matrix (5a) shows percentages with the predicted classes (normalized by column), so each column is a reflection of how accurate the model's predicted label is. When the model predicts Positive, 87.4% of those predictions correspond to true improvement years (true positives), while 12.6% correspond to false alarms (false positives). In contrast, when the model predicts Negative, 76.6% of those predictions are correct non-improvement years (true negatives), but 23.4% are missed improvement years. The ROC curve (5b) shows a moderate level of discriminative ability, with an AUC of 0.71. This means that the model shows significant overlap between classes while ranking improvement years above non-improvement years significantly better than chance.

Table 2 is a comparison of directional performance across all models by evaluating whether each approach correctly predicts the sign of annual food security change. While the earlier results focus exclusively on the Random Forest Δ FSI model for detailed diagnostics, Table 2 showcases the other models we tried by contextualizing how alternative methods and feature sets perform on the same 2013 to 2023 test window.

The best-performing model is the Random Forest using the full feature set (crops+macro+shocks+lags), achieving balanced accuracy = 0.8391 with high sensitivity

(0.8949) and strong specificity (0.7831), indicating reliable identification of both improvement and non-improvement years. Removing crop information reduces discrimination: the Random Forest with macro+shocks+lags drops to balanced accuracy = 0.7644 (sensitivity = 0.7458, specificity = 0.7831), suggesting that crop indicators provide additional signal beyond macro-climate shocks alone. Gradient Boosting exhibits an imbalanced error profile (balanced accuracy = 0.7085) with high sensitivity (0.8949) but lower specificity (0.5221), while RidgeCV further amplifies this trade-off by predicting positives aggressively (sensitivity = 1.0000) at the cost of very low specificity (0.2610) and lower balanced accuracy (0.6526). The persistence baseline performs near chance (balanced accuracy = 0.5221), meaning improvements not explained by persistence alone are captured most effectively by the full-feature Random Forest.

Figure 5: Confusion Matrix and ROC Curve for Predicting $\Delta\text{FSI} > 0$

Table 2: Directional Model Comparison for ΔFSI : Balanced Accuracy, Sensitivity, and Specificity (Test: 2013–2023)

Setup	Model	Balanced Accuracy	Sensitivity	Specificity
ΔFSI : crops+macro+shocks+lags	RandomForest	0.8391	0.8949	0.7831
ΔFSI : macro+shocks+lags	RandomForest	0.7644	0.7458	0.7831
ΔFSI : crops+shocks+lags	GradientBoosting	0.7085	0.8949	0.5221
ΔFSI : crops+shocks+lags	RidgeCV	0.6526	1.0000	0.2610
Baseline	Persistence ($\Delta=0$)0.5	0	1	

4 Discussion

Overall, our results show that national crop yield, production, and harvest-area statistics can help track US food security dynamics over time. The Random Forest ΔFSI framework provided the most reliable performance among the tested approaches, demonstrating that supervised models can learn relationships between crop indicators and a composite food security index using annual U.S.-only data (1970–2023), particularly when the target is modeled as YoY change (ΔFSI) rather than absolute levels. In the 2013–2023 period, the reconstructed predictions generally followed the observed trajectory and performed competitively against the persistence benchmark, supporting that national crop conditions correspond to broader system outcomes and can contribute to monitoring and forecasting.

A key takeaway is that shock and lag feature engineering impacts as much as model choice. The shock-lag sensitivity results show that removing both YoY shock terms and lag structure worsened performance, whereas including both consistently improved results. This is possible in food systems where production outcomes, costs, and price conditions influence availability and stability with delays rather than instantaneously. Consistent with this mechanism, lagged crop quantities (e.g., $t-1$, $t-2$) and YoY shock terms (%) appear among the strongest predictors in Figure 4, suggesting that the model benefits from both the “memory” of prior conditions and the detection of sudden disruptions.

Meanwhile, our results clarify what production-centered prediction can and cannot do. The residual analysis shows that the largest errors occur around sharper transitions, indicating that turning points are harder to reconstruct from crop and engineered macro-climate features. Directional up/down changes in ΔFSI shows moderate discrimination ($\text{AUC} \approx 0.71$), which is consistent with food security not being determined by production alone; affordability, distribution, macroeconomic shocks, and policy/trade dynamics can shift outcomes even when it is a harvest year. As a result, production and commodity indicators are most useful for identifying medium-term trends and gradual movement, and less trustworthy when other factors dominate the signal.

Several factors also limit directional performance. First, the test window spans only 11 years, so a few borderline years could shift the ROC curve. Second, the positive/negative boundary is susceptible to small noise in either the predictors or the constructed FSI, because many national YoY changes are small and near $\Delta\text{FSI} = 0$. Third, even when agricultural production contains signal, the sign of YoY change depends on drivers only partially represented in our features (e.g., policy shifts, trade dynamics, household prices, and access constraints). For these reasons, the ROC/AUC results should be interpreted as a supporting diagnostic indicating some ability to detect improvement years rather than a definitive early-warning classifier.

A further limitation concerns the binary directional definition itself (Positive if $\Delta\text{FSI} > 0$, Negative if $\Delta\text{FSI} \leq 0$). The sign alone can overstate practical significance: for example, if FSI rises from 0.60 to 1.00 and then decreases slightly to 0.99, the final year is labeled “Negative” despite remaining substantially more food secure than the initial year. Future work could introduce a magnitude-based warning rule (e.g., only flagging declines when the relative drop exceeds a threshold such as $\%\Delta\text{FSI} \leq -10\%$) or a neutral band around zero to separate fluctuations from policy-relevant deterioration.

Additional limitations should be considered. Although FSI provides a consistent 0–1 target, any index depends on design choices that can affect YoY dynamics. Despite including several macro and commodity-price series, the feature set omits important determinants such as household income and poverty, policy and assistance changes, trade volumes, inventories/stock-to-use conditions, and distribution constraints.

Otherwise, our findings are directly relevant to policy. First, they support using available crop statistics as part of an early-warning style monitoring toolkit, particularly for detecting deviations from recent norms when using YoY shocks and short lags. This can help agencies prioritize attention, scenario planning, and preparedness. Second, the limitations are equally informative: moderate ROC/AUC performance and outlier years reinforce that supply-side strength alone does not guarantee improved food security, implying that yield-boosting or acreage-expansion policies should be paired with measures targeting affordability and access (e.g., price stabilization).

5 Conclusion

Even in a high-producing agricultural system, such as US, national food security can also change rapidly, making timely monitoring and early warnings essential. In conclusion, crop indicators by themselves cannot fully explain food security, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The key finding from this work is that machine learning model is a promising approach to policy-relevant forecasting, and that year to year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects. Future work should expand our predictor set to include more variables such as access, trade, and policy drivers and move toward uncertainty-aware forecasts. Doing so allows more effective allocation of resources and earlier responses to food-security risks.

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References

- [1] Michael A. Long, Lara Gonçalves, Paul B. Stretesky, and Margaret Anne Defeyter. Food insecurity in advanced capitalist nations: A review. *Sustainability*, 12(9), 2020.
- [2] Aikaterini Kavallari, Thomas Fellmann, and S. Hubertus Gay. Shocks in economic growth = shocking effects for food security? *Food Security*, 6(4):567–583, August 2014.
- [3] Robert H. Beach, Allison M. Thomson, and Bruce A. McCarl. Climate change impacts on us agriculture. IATRC Public Trade Policy Research and Analysis Symposium, 2010. Available from: <https://ageconsearch.umn.edu/record/91393>.
- [4] Regis Barnichon, Christian Matthes, and Alexander Ziegenbein. Are the effects of financial market disruptions big or small? *The Review of Economics and Statistics*, 104(3):557–570, 05 2022.
- [5] J. L. Hatfield, K. J. Boote, B. A. Kimball, L. H. Ziska, R. C. Izaurralde, D. Ort, A. M. Thomson, and D. Wolfe. Climate impacts on agriculture: Implications for crop production. *Agronomy Journal*, 103(2):351–370, 2011.
- [6] Jasmien De Winne and Gert Peersman. Macroeconomic effects of disruptions in global food commodity markets: Evidence for the united states. *Brookings Papers on Economic Activity*, 2016(2):183–286, 2016. Fall 2016.
- [7] Maayan Kreitzman, Eric Toensmeier, Kai M. A. Chan, Sean Smukler, and Navin Ramankutty. Perennial staple crops: Yields, distribution, and nutrition in the global food system. *Frontiers in Sustainable Food Systems*, Volume 4 - 2020, 2020.
- [8] N. M. A. Manap and N. W. Ismail. Food security and economic growth. *International Journal of Modern Trends in Social Sciences*, 2(8):108–118, 2019.
- [9] Noé J. Nava, William Ridley, and Sandy Dall’erba. A model of the u.s. food system: What are the determinants of the state vulnerabilities to production shocks and supply chain disruptions? *Agricultural Economics*, 54(1):95–109, 2023.
- [10] B. Gail Smith. Developing sustainable food supply chains. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 363(1492):849–861, 08 2007.

- [11] Economist Impact Expert panel. Global food security index 2022: Exploring challenges and developing solutions for food security across 113 countries, 2022. Available from: <https://impact.economist.com/sustainability/project/food-security-index/>.
- [12] Akinniyi Akinwande. Nigeria agri-econ indicators (1950–2023), 2024. Available from: <https://www.kaggle.com/datasets/akinniyiakinwande/nigeria-agri-econ-indicators-1950-2023>.
- [13] Farhana Bibi and Azizur Rahman. An overview of climate change impacts on agriculture and their mitigation strategies. *Agriculture*, 13(8), 2023.
- [14] Rosana Salles-Costa, Aline Alves Ferreira, Ruben Araujo de Mattos, Michael E. Reichenheim, Rafael Pérez-Escamilla, Juliana de Bem-Lignani, and Ana Maria Segall-Corrêa. National trends and disparities in severe food insecurity in Brazil between 2004 and 2018. *Current Developments in Nutrition*, 6(4), Apr 2022.
- [15] FAOSTAT. Country indicators, 2025. Available from: <https://www.fao.org/faostat/en/#country>.
- [16] M Nubé. Confronting dietary energy supply with anthropometry in the assessment of undernutrition prevalence at the level of countries. *World Development*, 29(7):1275–1289, 2001.
- [17] Maeve Henchion, Maria Hayes, Anne Maria Mullen, Mark Fenelon, and Brijesh Tiwari. Future protein supply and demand: Strategies and factors influencing a sustainable equilibrium. *Foods*, 6(7), 2017.
- [18] Peter Warr. Food insecurity and its determinants. *Australian Journal of Agricultural and Resource Economics*, 58(4):519–537, 2014.
- [19] Ibrahim Bozkurt and Muhammed Veysel Kaya. Agricultural production index: International comparison. *Agricultural Economics (Czech Republic)*, pages 236–245, 2021.
- [20] U.S. Department of Agriculture, National Agricultural Statistics Service. Quick Stats Agricultural Database, 2025. Accessed via the Quick Stats API: <https://quickstats.nass.usda.gov/api>.
- [21] Varshita Nalluri. Crop price prediction dataset, 2024. Available from: <https://www.kaggle.com/datasets/varshitanalluri/crop-price-prediction-dataset>.
- [22] World Bank Group and partners. World development indicators - databank, 2015. Available from: <https://databank.worldbank.org/id/db4738db>.
- [23] World Bank Group and partners. World bank solutions to food insecurity - food security update, 2025. Available from: <https://www.worldbank.org/en/topic/agriculture/brief/food-security-update>.
- [24] Justin Oh. Total emissions per country (2000–2020), 2023. Available from: <https://www.kaggle.com/dsv/5097715>.
- [25] Hannah Ritchie, Pablo Rosado, and Max Roser. Agricultural production. *Our World in Data*, 2023. Available from: <https://ourworldindata.org/agricultural-production>.
- [26] Python Software Foundation. Python Language Reference, version 3.11, Oct 2024. <https://docs.python.org/3/reference/index.html>.
- [27] The pandas development team. *pandas-dev/pandas: Pandas*, February 2020.
- [28] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, and Ralf Gommers et al. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020.
- [29] Python Software Foundation. The Python Programming Language. Documentation 3.11, 2026. Specifically for the pathlib module.

[30] F. Pedregosa, Gaël Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al. Scikit-learn: Machine learning in python. *Journal of machine learning research*, 12(Oct):2825–2830, 2011.

Sources

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LEVERAGING BIG DATA & AI TO STRENGTHEN FOOD SECURITY & PROMOTE SUSTAINABLE AGRICULTURE

USDA FAS API Key:

j3sm6atK8zzEDndjK7dclUPem55DtFBf1IifSt0g

<https://apps.fas.usda.gov/opendatawebV2/#/home>

USDA QuickStats API Key:

6F514F01-32FC-397B-B0CE-78B7857D8A9E

https://github.com/cats-codes/HSBDC_202526

https://app.uio.no/cgi-bin/ifi/texcount/texcount_3_2_0_41.cgi

https://ers.usda.gov/sites/default/files/_laserfiche/outlooks/37116/31029_gfa14-b_002.pdf?v=21160

 2025-26 HSBDC Manuscript Guidelines

USDA CHARTS AND GRAPHS: https://www.nass.usda.gov/Charts_and_Maps/

- Crop progress/condition: https://www.nass.usda.gov/Research_and_Science/Crop_Progress_Gridded_Layers/index.php
- CDL map: <https://croplandcros.scinet.usda.gov/> ** idk if theres anyway to make it into raw numbers data.

General Sources/Links:

- Cats manuscript from last year:  cat's manuscript >:3 (its mine now)(
- [Goal 2 | Department of Economic and Social Affairs](#) (this is what we are trying to find a solution for)
- <https://databank.worldbank.org/id/9a9ceb98>
- [Google Colab \(For Data stuff\)](#)

- [Free Vancouver Citation Generator \[Updated for 2025\]](#)
- <https://www.fao.org/faostat/en/#home>

- For FAO:
 - These stats are **3-year averages** and go back to **2000**
 - People Undernourished and Prevalence of Undernourishment good
 - Food Availability also there (**Average dietary energy supply adequacy!!!**)
 - Number of moderately and severely food insecure people and Prevalence of moderate and severe food insecurity (%) go back to **2014**
 - Brazil: <https://www.fao.org/faostat/en/#country/21>
 - People Undernourished and Prevalence of Undernourishment have **gaps from 2012-2020**
 - Nigeria: <https://www.fao.org/faostat/en/#country/159>
 - Same as above with no gaps in data!
 - France: <https://www.fao.org/faostat/en/#country/68>
 - People Undernourished and Prevalence of Undernourishment **unavailable**
 - India: <https://www.fao.org/faostat/en/#country/100>
 - Number of moderately and severely food insecure people and Prevalence of moderate and severe food insecurity (%) **unavailable**
 - USA: <https://www.fao.org/faostat/en/#country/231>
 - People Undernourished and Prevalence of Undernourishment **unavailable**
 - Actually usable is
 - Average dietary energy supply adequacy (2000, 3 years)
 - Per capita food supply variability (2001, yearly)
 - (Agricultural Area and Land Usage if relevant?)
 - GDP per capita (see below)

- [Food insecurity across age: Evidence from a global study - ScienceDirect](#)
 - Definitions of food insecurity and its dangers
 - 2014-2023
 - Food security across different stages of life
 - Long-term trends as well as disruptions (COVID)

- [National Trends and Disparities in Severe Food Insecurity in Brazil between 2004 and 2018 - PMC](#)
 - trends in food insecurity in Brazil from 2004-2018

- [The State of Food Security and Nutrition in the World 2025](#)
 - this data is all very recent so may not be very useful
 - food insecurity is impacted by inflation
 - food security and nutrition:
 - more recent updates, signs of some improvement in some areas
 - basically just the amounts of food insecurity in different areas
 - affordability of healthy diets (don't know how relevant that is)

- causes and consequences for food security and nutrition
- [KPIs for Food Waste, Insecurity, & Overconsumption | Kaggle](#)
 - defining food security and insecurity, other really useful definitions
 - social aspect of climate change
 - food waste per capita nutrition loss; food loss per food-insecure population; meat overconsumption
- [Food Security Update | World Bank Solutions to Food Insecurity](#)
 - again this is very recent data but could give us some information about:
 - how the World Bank addresses food insecurity (possible tactics to deal with food insecurity)
 - ways that climate change impacts agricultural practices

Socioeconomic:

- FAOstat (see above) has GDP per Capita for all the countries (2000, yearly)
- [Crop Price Prediction Dataset](#) - linked with agriculture
- [NIGERIA INFLATION RATES](#)
 - 1.iamHardy. NIGERIA INFLATION RATES [Internet]. Kaggle.com. 2024 [cited 2025 Nov 30]. Available from: <https://www.kaggle.com/datasets/iamhardy/nigeria-inflation-rates>
- [Nigeria Agri-Econ Indicators \(1950-2023\)](#)
 - 1.Akinwande A. Nigeria Agri-Econ Indicators (1950-2023) [Internet]. Kaggle.com. 2023. Available from: <https://www.kaggle.com/datasets/akinniyiakinwande/nigeria-agri-econ-indicators-1950-2023>
- [Human Development Index](#)
 - 1.United Nations Development Programme. Human Development Index [Internet]. United Nations Development Programme. United Nations; 2024. Available from: <https://hdr.undp.org/data-center/human-development-index#/indicies/HDI>

Agricultural Production:

- <https://www.kaggle.com/code/abhijitdahatonde/crop-production-trends>
 - 1.abhijitdahatonde. 🌽🌾 Crop Production Trends 📈🌾 [Internet]. Kaggle.com. Kaggle; 2024 [cited 2025 Nov 30]. Available from: <https://www.kaggle.com/code/abhijitdahatonde/crop-production-trends>
- <https://www.kaggle.com/code/jacopoferretti/exploring-the-agricultural-production-in-the-usa>
 - 1.jacopoferretti. Exploring the Agricultural Production in the USA [Internet]. Kaggle.com. Kaggle; 2025. Available from: <https://www.kaggle.com/code/jacopoferretti/exploring-the-agricultural-production-in-the-usa>
- [Agricultural Production - Our World in Data](#)
 - 1.Ritchie H, Roser M, Rosado P. Agricultural Production. Our World in Data [Internet]. 2023; Available from: <https://ourworldindata.org/agricultural-production>
- [India Crop Production - State wise](#)
- [Brazilian Agricultural Production](#)

- [European Agriculture Indicators](#)
- [India at a glance](#)
- [Food Trade Dependence Index | Tool](#) - this is missing some of the countries
- [Cereal import dependency ratio - Table - Food Systems Dashboard](#)

Emissions and Sustainability:

- [Total Emissions Per Country \(2000-2020\)](#)
- [Methane Emissions due to Rice crop](#)
 - data from FAOstat
 - 1961-2020
 - a lot of different countries
 - different types of emissions and from different things
- [Environment | Data](#)

Climate Change:

- [Public EM-DAT](#)
 - Contains records of history of natural disaster across different countries
- [Historical Data and Conditions | Drought.gov](#)
- [A global dataset for the projected impacts of climate change on four major crops | Scientific Data](#)
 - 1.Hasegawa T, Wakatsuki H, Ju H, Vyas S, Nelson GC, Farrell A, et al. A global dataset for the projected impacts of climate change on four major crops. Scientific Data. 2022 Feb 16;9(1).
- [Climate change cuts global crop yields, even when farmers adapt | Stanford Doerr School of Sustainability](#)
 - 1.Garthwaite J. Climate change cuts global crop yields, even when farmers adapt [Internet]. Stanford Doerr School of Sustainability. Stanford University; 2025. Available from: <https://sustainability.stanford.edu/news/climate-change-cuts-global-crop-yields-even-when-farmers-adapt>
- [What Is Climate Change? | United Nations](#)
 - 1.United Nations. What is Climate Change? [Internet]. United Nations. 2025. Available from: <https://www.un.org/en/climatechange/what-is-climate-change>
- [The socioeconomic impact of climate change in developing countries over the next decades: A literature survey - ScienceDirect](#) - ALSO SOCIOECONOMIC
 - 1.Adom PK. The socioeconomic impact of climate change in developing countries over the next decades: A literature survey. Heliyon. 2024 Aug 1;10(15):e35134–4.
 - [Climate Change & Agriculture Analysis | Kaggle](#)
 - [Other Global Soil Maps and Databases | FAO SOILS PORTAL | Food and Agriculture Organization of the United Nations](#)
 - [spei.csic.es](#) - drought severity
 - https://cds.climate.copernicus.eu/datasets/derived-drought-historical-monthly?tab=overview&utm_source=chatgpt.com - rainfall (SPI for rainfall only)
 - <https://ourworldindata.org/grapher/country-level-monthly-temperature-anomalies> - temperature anomalies

— [Indian Agriculture and Climate Dataset \(1961-2018\)](#) — linked with agriculture

Food Waste:

- [Food Waste | Data Import & EDA Starter](#)
 - 1.joebeachcapital. Food Waste | Data Import & EDA Starter [Internet]. Kaggle.com. Kaggle; 2024. Available from:
<https://www.kaggle.com/code/joebeachcapital/food-waste-data-import-eda-starter/input>
- [Food Waste | Kaggle](#)
 - 1.Food Waste [Internet]. www.kaggle.com. Available from:
<https://www.kaggle.com/datasets/joebeachcapital/food-waste/data>

Agricultural Practices of India:

[Different Types of Agriculture in India: Top Farming Methods.](#)

https://mospi.gov.in/sites/default/files/reports_and_publication/cso_social_statices_division/comp_e nv_2008-09_chp5.pdf

[INDIAN LAND DEGRADATION DATA SET](#)

- Heavy fertilizer use (causes soil degradation)
- Subsistence farming
- Extensive farming
- Plantation farming
- Commercial farming
- Dryland farming
- Shifting cultivation
- Capitalist farming
- Mixed farming
- Mediterranean farming
- Primary crops: rice, wheat, corn, millet, rapeseed, soybean, peanut, legumes/pulses

Agricultural Practices of Brazil:

- Primary crops: coffee, sugar, soybeans, manioc, rice, maize, cotton, edible beans and wheat
- Irrigation: (THE SOURCES ARE ALL FROM 2008!!!!)

Agricultural Practices of France:

- Primary crops: wheat, corn, sugar beets, and potatoes

Agricultural Practices of Nigeria:

-

Agricultural Practices of the USA:

-

Agricultural Practices Consistent Across All Countries:

Plan of Attack

Plan of Attack:

- Measuring climate change via soil degradation, rainfall, etc; [factor] over time per country; visualizing the data itself
- Then find the correlation between said factors with agricultural practices: if the practice goes up in response to climate change vs. if the practice stays the same despite climate change, how will this affect food security in the next 5 years
 - Ex. because more irrigation is used, the food scarcity in the next 5 years is predicted to decrease; and if irrigation is kept constant then what will happen to food scarcity in the next 5 years in relation to soil degradation/rainfall amount
- Do this for each country for each practice and then recommend which practice to invest most in based on our findings
 - Ex. use more fertilizer, more irrigation, import or whatever

In order to keep climate change accounted for, agricultural practices and defining factors for climate change (e.g soil degradation, drought) **must be constant across all countries**

All the practices are tested against each country's food scarcity. The rise in practices will be predicted off of the climate change factors.

Then compare which practice decreases each country's food scarcity the most, we recommend the said practice for said country

1. visualizing the climate change factors on a graph
2. visualizing current agricultural practices on a graph too
3. predicting future agricultural practices
4. visualizing current food scarcity trend for each country
5. taking futural agricultural practice predictions and in turn using that to predict future food scarcity trends
6. look at which practice affects each country's food scarcity in the best way the most
7. recommending said practice in the discussion and give ideas as to why each country has the practice that it does have

What we need to do:

- Figure out factors for measuring climate change, like soil degradation, natural disasters? etc. (how many should we have?)
 - List so far:
 - Soil degradation
 - Natural Disasters frequency
 - Flood
 - Drought ** primary one
 - Severe Storms
 - Temperature → also look at like... extreme temperature

- Rainfall
- Growing Season?
- Figure out all the agricultural practices in each country (irrigation, fertilization, etc.)

Limitations:

- Not examining by regions per country
- Data is imperfect (some gaps)

QUESTIONS:

- What to do with data sets representing the same metric from different sources (if they differ)
 - Eg. World Hunger Map vs FAOstat hunger
 - ok
 - as long as its the same SCALE (eg. per capita/per 100k people), it doesnt matter bc all the data is congregated by the unit so it doesnt matter if its from different sources or not
- Time range?
 - as much as possible. i saw that we cant to predict like eg. 10 years into the future so the more data we have to create a decent history to go off of, the more we can predict
 - also all data needs to be in the same unit or else they cant be combined
- Figure out focused research question
 - 🤔 this can also be reworded based on how we are modeling it and what results we get lmfao
 - but jsut have a generalized idea first
- Figure out methods how do to project
 - ok what we're doing is acc really similar to my youreka project from earlier this year, just a different concept
 - for that one we used generalized linear models :)) so we might be able to use the same model
- How are we building the model?
 - python my love
- Since usa, india, brazil, france, nigeria are some of the largest countries in the world (ya know, land and people) are we going to have to divide it up by region and such and such? Or like when testing/training, should we focus on sample regions, then let the AI handle the country level?
 - no treat each country as its independent thing without splitting it further. otherwise, it gets wayy too complicated
- How define food security/insecurity
 - uhhhhmmmm look at the data/literature review

Issues we are finding:

- Hard to find what practices are used across all countries (sources are old)
- Are we dealing with too much data

One for climate change factors

One for oractices

Compare rise and fall

Find common agricultural practices (can even be numbers on a graph)

Stuff to define:

- Food insecurity
- Climate change (effects of climate change as variables)
- Agricultural practice (variables)

chatgpt

Here's a clean way to tighten your plan so it's *doable*, consistent across countries, and still answers the “what to invest in” question.

1) Lock in a single food-security target (don't mix)

Pick **one** outcome for the whole project:

- **SDG 2.1.2 (FIES): % of population moderately or severely food insecure**
Pros: directly measures access-to-food experience; widely used; comparable across countries. ([FAOHome](#))
- Alternative: **SDG 2.1.1 (PoU): prevalence of undernourishment (%)**
Pros: classic “hunger” metric; Cons: model-based and smoother/less responsive year-to-year. ([FAOHome](#))

If you want 5-year forecasting + policy responsiveness, **FIES (2.1.2)** is usually the better fit.

2) Keep climate-change “factors” small (3–5) and defensible

You don't need 10 climate variables. You need a compact set that:

- exists globally,
- is annual (or can be aggregated annually),
- isn't redundant.

Recommended set (works for almost all countries):

1. **Drought severity** (primary): use **SPEI** (includes temperature effects) ([spei.csic.es](#))
2. **Rainfall / precipitation anomaly** (or SPI if you prefer precipitation-only; often bundled with drought products) ([Climate Data Store](#))
3. **Temperature anomaly / extreme heat proxy** (annual mean anomaly or “extreme temp days” if you can get it consistently)
4. **Disaster frequency** (optional, but nice): natural hazard events per year from **EM-DAT** ([emdat.be](#))
5. **Soil degradation** is great *conceptually* but is the hardest to get as a consistent annual country series. If you can't find it annually, treat it as:
 - a slower-moving “baseline” risk index (static/low frequency), or
 - drop it and use land productivity / vegetation proxies instead.

Rule of thumb: start with (**drought + rainfall + temperature + disasters**). Add soil only if you can get a consistent series.

3) Pick agricultural practices that actually exist as comparable country-year data

Your biggest pain point (“sources are old / inconsistent”) goes away if you choose practices with strong global coverage.

High-coverage practice candidates:

- **Irrigation:** % of agricultural land irrigated (World Bank) ([World Bank Open Data](#))
- **Fertilizer intensity:** kg per hectare of arable land (World Bank) ([DataBank](#))
- **(Optional) Imports:** use a *food import dependency* / *net food imports* type variable if you can source it consistently (coverage varies—only include if it’s solid for your country set)

If you try to include “improved seed adoption,” “precision ag,” etc., you’ll drown in missing data.

4) The modeling move that makes your counterfactuals valid-ish

What you’re describing is basically:

food_insecurity(t) = f(climate(t), practices(t), interactions, controls)

Key idea: include **interaction terms** so you can say “when drought worsens, irrigation matters more.”

Example (panel regression / GLM idea):

- Outcome: **FIES_it** (or **PoU_it**)
- Predictors:
 - climate: **SPEI_it**, **rain_anom_it**, **temp_anom_it**, **disasters_it**
 - practices: **irrig_it**, **fert_it**
 - interactions: **SPEI_it * irrig_it**, **SPEI_it * fert_it**
 - plus a few controls if you can (GDP per capita, conflict proxy, etc.—optional but helps)

Then you can run two scenarios per country:

- **Adaptive:** practices follow the predicted response to climate
- **Constant-practice:** hold practice at last observed level and let climate change

5) Your “same metric from different sources” issue (do this instead)

Don't "average" hunger metrics from different sources. Choose **one authoritative definition** and stick to it.

Good approach:

- **Food insecurity:** FAO SDG series (2.1.2 or 2.1.1) ([FAOHome](#))
- **Disasters:** EM-DAT only ([emdat.be](#))
- **Fertilizer / irrigation:** World Bank WDI indicators ([DataBank](#))

If two sources claim the "same" metric but disagree:

- prefer the one with clearer metadata + consistent update cadence,
- and document the choice as a limitation (measurement differences).

6) Time range

For 5-year forecasting, prioritize **recent + consistent**:

- Aim for **~2000–2024** if available annually.
- If a variable only starts in 2001–2021 (common for some WDI series), align everything to the **overlap window** and move on.

More years helps, but *only if coverage stays consistent*.

7) Are you dealing with too much data?

Right now, yes—because you're trying to do:

- too many climate factors,
- too many practices,
- and maybe too many countries.

A safe, strong scope:

- **5 countries × 2 practices × 3–4 climate factors × 1 outcome**
 - annual data, 20-ish years
- This is already plenty for clear results + visuals + discussion.

8) Concrete "next steps" checklist (so you can start coding)

1. Choose outcome: **FIES (2.1.2)** or **PoU (2.1.1)** ([FAOHome](#))

2. Climate set: **SPEI + precip anomaly + temp anomaly + (optional) EM-DAT disasters** ([Climate Data Store](#))
3. Practices set: **irrigation + fertilizer** ([World Bank Open Data](#))
4. Build a single country-year panel table (merge on ISO3 + year)
5. Fit model with interactions
6. Forecast climate (or take exogenous projections if you have them), forecast practices, then forecast food insecurity
7. Run counterfactual: constant-practice vs adaptive-practice
8. Rank practices by **predicted reduction** in food insecurity for each country

If you want, paste your draft list of target countries + which food-security metric you prefer (FIES vs PoU), and I'll lay out an exact dataframe schema + the model formulas to implement in Python.

Idealization

Here is a prototype ML design for **spring wheat zone recommendations** targeting **Alberta (AB)** or **Minnesota (MN)**, using **PyTorch + QGIS** as the GeoAI stack. **QGIS/DB Manager/GDAL/QGIS Server/qgis2web** are utilized for GeoAI tasks.

1) Scope & assumptions

- **Crop:** *Spring wheat* (common in AB and MN).
 - **Spatial scale:** Field or 30 m grid derived from annual crop masks (ACI, AAFC Annual Crop Inventory, in Canada, CDL, Cropland Data Layer, in U.S.). ([open.canada.ca](#)), ([nass.usda.gov](#))
 - **Goal:** Recommend **variable-rate inputs** (seeding, N timing; irrigation only where present) using **predicted yield** and **water-use efficiency (WUE = Yield/ET)**.
 - **Auditability:** Aggregate predictions back to **StatCan** (AB) and **NASS QuickStats** (MN) to ensure consistency with official totals. ([www150.statcan.gc.ca](#)), ([nass.usda.gov](#))
-

2) Data architecture (AB vs MN)

Alberta (Canada)

- **Crop mask: AAFC Annual Crop Inventory (ACI)** 30 m crop type rasters via Open Government, REST/WCS, and Earth Engine; use wheat class by year. [open.canada.ca], [[developers...google.com](https://developers.google.com/)]
- **Agroclimate/heat: AAFC climate normals** (GDD, Crop Heat Units, frost-free period) for prairie phenology features. [agriculture...canada.ca]
- **Drought & soil moisture: Canadian Drought Monitor (CDM)** monthly classes (D0–D4) and AAFC **Drought Watch** indices (SPI/SPEI/soil moisture). [agriculture...canada.ca], [agriculture...canada.ca]
- **Vegetation time series:** NDVI composites (e.g., StatCan CCAP NDVI or GEE) for seasonal curves. [statcan.gc.ca]
- **ET for WUE: MODIS MOD16A2** (8-day, 500 m) ET via Earth Engine; use MOD16A2GF where needed. [[developers...google.com](https://developers.google.com/)], [lpdaac.usgs.gov]
- **Audit labels: Statistics Canada** principal field crop area/production/yields—auditing at province/regional level. [www150.statcan.gc.ca]

Minnesota (U.S.)

- **Crop mask: NASS Cropland Data Layer (CDL)** annual crop types (CroplandCROS). [nass.usda.gov]
- **Vegetation: VegScape** NDVI/condition (weekly). [nass.usda.gov]
- **ET for WUE: OpenET**—daily/monthly ET covering the **contiguous 48 U.S. states** (Dec 16, 2025 expansion). [etdata.org]
- **Audit labels: USDA NASS QuickStats** county wheat yield/production via API/JSON. [nass.usda.gov]

Cross-border note: WUE in AB uses **MODIS ET**; in MN, **OpenET**. Document method differences when comparing WUE across regions. [[developers...google.com](https://developers.google.com/)], [etdata.org]

3) Feature schema (per field/30 m cell)

Temporal sequence (weekly):

- NDVI curve features (green-up, peak, integral, senescence slope). AB: CCAP/GEE; MN: VegScape. [statcan.gc.ca], [nass.usda.gov]
- **Heat accumulation:** GDD (base 5/10 °C), CHU; AAFC normals in AB; derived climate inputs in MN. [agriculture...canada.ca]
- **Water stress:** ET deficit (ET – precip proxy), SPEI/SPI anomalies; AB via AAFC Drought Watch. [agriculture...canada.ca]

Static/context:

- Crop mask & rotation frequency: ACI/CDL multi-year sequences. [\[open.canada.ca\]](https://open.canada.ca/), [\[nass.usda.gov\]](https://nass.usda.gov/)
- Basic topography indices (slope/TWI from DEM).
- Drought exposure (AB): cumulative **CDM** weeks in D1–D3 during the season. [\[agriculture....canada.ca\]](https://agriculture....canada.ca/)

Targets (labels):

- **Downscaled field yield** via **dasymetric allocation** of official production to field cells using phenology/heat/stress weights; **audit** back to province (AB) or county (MN) totals. [\[www150.statcan.gc.ca\]](https://www150.statcan.gc.ca/), [\[nass.usda.gov\]](https://nass.usda.gov/)
-

4) Model architecture (PyTorch)

- **Backbone:** **Temporal Transformer** (or **ConvLSTM**) over weekly sequences (NDVI, GDD/CHU, ET, stress), concatenated with static features.
 - **Heads:**
 - **Yield** (t/ha or bu/ac)
 - **WUE** (auxiliary regression or derived as yield / ET)
 - **Audit loss:** Penalize mismatch between **aggregated predictions** and **official totals** (province for AB, county for MN). [\[www150.statcan.gc.ca\]](https://www150.statcan.gc.ca/), [\[nass.usda.gov\]](https://nass.usda.gov/)
 - **Loss:** $L = \text{RMSE}_{\text{yield}} + \lambda \cdot \text{AuditLoss} + \gamma \cdot \text{SmoothnessLoss}$
 - **CV:** Spatial CV (leave-region/county out) + Temporal CV (train previous years, test latest).
 - Spatial cross-validation (leave-county/leave-state out) to avoid spatial leakage.
 - Temporal CV (train on 2012–2019, test 2020–2024).
-

5) Zone recommendation logic (seasonal)

Define **three zone classes** per field/grid:

1. **High-Potential / High-WUE** → consider **higher seeding rate** or **optimized N timing**; maintain conservative irrigation where applicable.
2. **High ET / Low-WUE** → reduce irrigation (if irrigated), adopt **split N** or adjust timing; consider drought-tolerant varieties.
3. **Low Potential / Stable ET** → maintain baseline inputs; focus on stand establishment and residue management.

Rules combine **percentiles** for **Yield_pred**, **WUE**, **NDVI integral**, plus **CDM exposure in AB** or **ET anomaly in MN** to constrain recommendations (practical precision-ag logic).

[\[agriculture....canada.ca\]](https://agriculture....canada.ca/), [\[etdata.org\]](https://etdata.org/)

6) QGIS-based GeoAI pipeline

6.1 Ingest, manage & automate (QGIS + DB Manager + GDAL)

- Use **QGIS Processing framework** to chain steps (clip AOI, mask crops, compute zonal features) via the **Model Designer** for repeatable workflows and batch runs. [\[docs.qgis.org\]](#), [\[docs.qgis.org\]](#)
- Store features and outputs in **PostGIS** or **GeoPackage** via **DB Manager**, including SQL queries, spatial indexing, and table management. [\[docs.qgis.org\]](#), [\[docs.qgis.org\]](#)
- Apply **GDAL** tools for raster ETL—[gdalwarp](#) to reproject/resample ET/NDVI rasters, [gdal_calc](#) for raster math (e.g., ET deficits). [\[gdal.org\]](#), [\[gdal.org\]](#)

6.2 Temporal visualization

- Use the **Temporal Controller** (built into QGIS ≥3.14) to animate weekly NDVI/ET features or show seasonal zone evolution; TimeManager plugin is legacy. [\[qgistutorials.com\]](#), [\[anitagraser.com\]](#)

6.3 Web publishing

- **QGIS Server** to publish **WMS/WFS/WMTS** & OGC API Features (for live services to ops dashboards). [\[docs.qgis.org\]](#), [\[github.com\]](#)
 - Rapid sharing via **qgis2web** (Leaflet/OpenLayers) to export interactive web maps directly from your QGIS project (no server-side required). [\[plugins.qgis.org\]](#), [\[github.com\]](#)
-

7) Example project configs

7.1 Alberta (AB) — spring wheat

YAML

project: ab_spring_wheat_zones

aoi: data/aoi/ab_boundaries.gpkg

years: [2019, 2020, 2021, 2022, 2023, 2024, 2025]

datasets:

crop_mask: AAFC_ACI_30m # wheat class via REST/WCS/GEE

[1](https://open.canada.ca/data/en/dataset/5489d54a-9925-44e3-a1a8-b26c1f64e811)[5](https://www.arcgis.com/home/item.html?id=aebe8247046d49f4b3220ef1e4e4640b)

ndvi: GEE_MODIS_S2_weekly # seasonal composites
[10](https://www.statcan.gc.ca/en/subjects-start/agriculture_and_food)
climate_normals: AAFC_GDD_CHU # normals
[7](https://agriculture.canada.ca/atlas/climate-normals-interactive-maps)
drought: Canadian_Drought_Monitor # monthly classes
[8](https://agriculture.canada.ca/en/agricultural-production/weather/canadian-drought-monitor)
et: MODIS_MOD16A2_500m # 8-day ET
[11](https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MOD16A2)
labels:
provider: Statistics_Canada # province totals (audit)
[3](https://www150.statcan.gc.ca/n1/daily-quotidien/250627/dq250627b-eng.htm)
features:
grid_size_m: 30
sequences: weekly
model:
type: temporal_transformer
loss_weights: {rmse: 1.0, audit: 0.3, smooth: 0.1}
outputs:
zones_gpkg: outputs/ab_wheat_zones.gpkg
publish_wms: qgis_server/ab_wheat_wms # QGIS Server
[22](https://docs.qgis.org/latest/en/docs/server_manual/index.html)
Show more lines

7.2 Minnesota (MN) — spring wheat

YAML

project: mn_spring_wheat_zones
aoi: data/aoi/mn_counties.gpkg
years: [2019, 2020, 2021, 2022, 2023, 2024, 2025]
datasets:
crop_mask: NASS_CDL # annual crop mask [2](https://www.nass.usda.gov/Data_and_Statistics/)
ndvi: VegScape_weekly # NDVI curves [2](https://www.nass.usda.gov/Data_and_Statistics/)
et: OpenET_API # daily/monthly ET (48 states)
[13](https://etdata.org/openet-expands-satellite-based-water-use-data-48-states-farmers-water-managers-across-nation/)
labels:
provider: NASS_QuickStats # county labels (audit) [4](https://www.nass.usda.gov/Quick_Stats/)
features:

grid_size_m: 30
sequences: weekly
model:
type: convlstm
loss_weights: {rmse: 1.0, audit: 0.5, smooth: 0.1}
outputs:
zones_gpkg: outputs/mn_wheat_zones.gpkg
web_export: web/mn_wheat_qgis2web # Leaflet export
[24](https://plugins.qgis.org/plugins/qgis2web/)
Show more lines

8) PyTorch prototype (temporal transformer + audit loss)

Python

model.py (simplified)

import torch, torch.nn as nn

```
class PositionalEncoding(nn.Module):
    def __init__(self, d_model, max_len=128):
        super().__init__()
        pe = torch.zeros(max_len, d_model)
        pos = torch.arange(0, max_len).unsqueeze(1)
        div = torch.exp(torch.arange(0, d_model, 2) * (-torch.log(torch.tensor(10000.0)) / d_model))
        pe[:, 0::2] = torch.sin(pos * div); pe[:, 1::2] = torch.cos(pos * div)
        self.register_buffer("pe", pe)
    def forward(self, x): # [B,T,D]
        return x + self.pe[:x.size(1)].unsqueeze(0)
```

```
class WheatTemporalTransformer(nn.Module):
    def __init__(self, seq_dim, static_dim, d_model=128, nhead=4, nlayers=3):
        super().__init__()
        self.proj = nn.Linear(seq_dim, d_model)
        enc_layer = nn.TransformerEncoderLayer(d_model=d_model, nhead=nhead, batch_first=True)
        self.encoder = nn.TransformerEncoder(enc_layer, nlayers)
        self.pe = PositionalEncoding(d_model)
        self.static_fc = nn.Sequential(nn.Linear(static_dim, d_model), nn.ReLU())
        self.head_yield = nn.Sequential(nn.Linear(2*d_model, 128), nn.ReLU(), nn.Linear(128, 1))
```

```

self.head_wue = nn.Sequential(nn.Linear(2*d_model, 128), nn.ReLU(), nn.Linear(128, 1))
def forward(self, seq_x, static_x):
    z = self.encoder(self.pe(self.proj(seq_x))) # [B,T,D] -> [B,T,D]
    z_t = z.mean(dim=1) # temporal pooling
    z_s = self.static_fc(static_x)
    h = torch.cat([z_t, z_s], dim=1)
    return self.head_yield(h).squeeze(-1), self.head_wue(h).squeeze(-1)

def audit_loss(preds, group_ids, official_totals: dict):
    # aggregate predictions to audit units (province/county)
    device = preds.device
    loss = torch.tensor(0.0, device=device)
    for gid, official in official_totals.items():
        mask = (group_ids == gid)
        loss = loss + (preds[mask].sum() - official)**2
    return loss / max(1, len(official_totals))

```

Show more lines

9) QGIS automation (PyQGIS + DB Manager + GDAL)

9.1 Ingest & feature build

Python

```

# qgis_etl.py (PyQGIS pseudocode)
from qgis.core import QgsProject, QgsRasterLayer, QgsVectorLayer
import processing # QGIS Processing framework
[14](https://docs.qgis.org/latest/en/docs/user_manual/processing/index.html)

# Load ACI (AB) or CDL (MN)
aci = QgsRasterLayer("url_to_AACI_2025_ImageServer", "ACI_2025") # wheat class in AB
[5](https://www.arcgis.com/home/item.html?id=aebe8247046d49f4b3220ef1e4e4640b)
cdl = QgsRasterLayer("/path/to/CDL_2025.tif", "CDL_2025") # MN crop mask
[2](https://www.nass.usda.gov/Data_and_Statistics/)

# Reproject ET rasters using GDAL (e.g., match AOI CRS)
processing.run("gdal:warp", { # gdalwarp utility
    "INPUT": "/path/modis_mod16a2_2025.tif", "TARGET_CRS": "EPSG:3347", "RESAMPLING": 0,

```

```
"OUTPUT": "/tmp/modis_et_ab_2025.tif"
}) # [18](https://gdal.org/en/stable/programs/gdalwarp.html)

# Build zonal features (mask wheat cells, sample NDVI/ET)
# ... use 'qgis:rastersampling', 'qgis:zonalstatistics', 'native:extractbyexpression'

# Write to PostGIS using DB Manager connections (or GeoPackage)
# DB Manager supports PostGIS/GeoPackage operations and SQL window
[16](https://docs.qgis.org/latest/en/docs/user_manual/plugins/core_plugins/plugins_db_manager.htm
l)
Show more lines
```

9.2 Temporal visualization & publishing

- **Temporal Controller** for seasonal animations of zones/NDVI. [\[qgistutorials.com\]](https://qgistutorials.com)
 - **QGIS Server**: publish project as **WMS/WFS/WMTS**; use [GetCapabilities](#), [GetMap](#), [GetFeatureInfo](#); optionally **GetPrint** for layouts/atlas. [\[docs.qgis.org\]](https://docs.qgis.org), [\[github.com\]](https://github.com)
 - **qgis2web**: export **Leaflet/OpenLayers** web map of zones ([yield_pred](#), [wue](#), [zone_class](#)) for quick stakeholder sharing. [\[plugins.qgis.org\]](https://plugins.qgis.org), [\[github.com\]](https://github.com)
-

10) Validation & audit dashboards (QGIS)

- **Spatial CV residual maps** (leave-county/province out) styled in QGIS.
 - **Audit panels**: Compare predicted aggregates vs **StatCan (AB)** and **NASS QuickStats (MN)**; include season filters via **Temporal Controller** and printable layouts via **GetPrint** (QGIS Server). [\[www150.statcan.gc.ca\]](https://www150.statcan.gc.ca), [\[nass.usda.gov\]](https://nass.usda.gov), [\[github.com\]](https://github.com)
-

11) 4–6 week implementation plan (QGIS-first)

1. **Week 1** — AOI & data staging in QGIS; Processing Model Designer skeleton; PostGIS schema via DB Manager. [\[docs.qgis.org\]](https://docs.qgis.org), [\[docs.qgis.org\]](https://docs.qgis.org)
2. **Week 2–3** — ETL for ACI/CDL, NDVI, ET, drought; **dasymetric allocation** baseline; **GBM** baseline (tabular) and export features as Parquet/NPY for PyTorch. [\[open.canada.ca\]](https://open.canada.ca), [\[nass.usda.gov\]](https://nass.usda.gov), [\[developers...google.com\]](https://developers.google.com), [\[agriculture...canada.ca\]](https://agriculture.canada.ca)
3. **Week 4** — Train PyTorch **temporal transformer** (add **audit loss**); Spatial/Temporal CV; SHAP/attention inspection.

4. **Week 5** — Generate **zone layers**; publish via **QGIS Server** and/or **qgis2web**; set up **Temporal Controller** animations. [\[docs.qgis.org\]](#), [\[plugins.qgis.org\]](#), [\[qgistutorials.com\]](#)
 5. **Week 6** — Stakeholder review; thresholds tuning; move to more counties/regions.
-

Why QGIS fits here

- **Processing Modeler** for chaining and rerunning complex spatial workflows (clip/mask/sample/zonal). [\[docs.qgis.org\]](#), [\[docs.qgis.org\]](#)
 - **DB Manager** for seamless **PostGIS** integration, spatial indexing, SQL analytics. [\[docs.qgis.org\]](#)
 - **GDAL** tools accessible in QGIS for reprojection/resampling/math. [\[gdal.org\]](#), [\[gdal.org\]](#)
 - **QGIS Server** for OGC services (WMS/WFS/WMTS, OGC API), and **qgis2web** for fast static web deployment. [\[docs.qgis.org\]](#), [\[github.com\]](#), [\[plugins.qgis.org\]](#)
-

Ready to proceed?

Tell me:

1. **Region choice**: **AB** or **MN** (or both).
2. **Geometry**: **30 m grid** (ACI/CDL native) or **field polygons** (if you have boundaries).
3. **Seasons**: default 2019–2025, or specify subset.

I'll deliver a **repo** with:

- **PyQGIS ETL scripts** (ACI/CDL, NDVI, ET, drought) and a **Processing Modeler** workflow,
- **PyTorch** training script (temporal transformer + **audit loss**),
- **QGIS Server** project and/or **qgis2web** export for **zone recommendations** with a simple audit panel—aligned to **StatCan/NASS** releases. [\[www150.statcan.gc.ca\]](#), [\[nass.usda.gov\]](#)

a **ready-to-run repository scaffold** for your AB + MN spring-wheat zone recommendations using **PyQGIS + PyTorch + QGIS Server + qgis2web** with **30 m** geometry and **2019–2025** seasons.

Repo path (created in this workspace): `./wheat-zones-geoai`

It contains configs, ETL templates, model code, QGIS project placeholders, and a Leaflet export template.

✓ What's inside (repo structure)

```
wheat-zones-geoai/
├── README.md
├── LICENSE
├── config/
│   ├── ab.yml          # AB setup (ACI, CDM, MODIS ET, AAFC normals)
│   └── mn.yml          # MN setup (CDL, VegScape NDVI, OpenET ET)
├── pyqgis/
│   ├── etl_ab.py       # PyQGIS ETL template for Alberta
│   └── etl_mn.py       # PyQGIS ETL template for Minnesota
├── training/
│   ├── train.py        # PyTorch training loop (Transformer, optional audit)
│   ├── models/
│   │   └── temporal_transformer.py
│   └── utils/
│       ├── dataset.py  # NPZ loader; expects seq/static/y/wue/group_ids/area
│       └── audit.py    # Optional audit loss helper
├── features/
│   ├── ab/             # ETL writes NPZ per year here (2019–2025)
│   └── mn/
├── qgis-project/
│   ├── ab_wheat_zones.qgs # Project for QGIS Server (load the GeoPackage output)
│   └── mn_wheat_zones.qgs
├── web/
│   └── index.html       # Leaflet template (qgis2web-style); load your GeoJSON
└── scripts/
    └── train.sh         # CLI helper to launch training
```

🚀 How to run (end-to-end)

1) ETL in QGIS (PyQGIS)

Open QGIS and run the ETL templates in the Python console:

- `pyqgis/etl_ab.py` (Alberta)
- `pyqgis/etl_mn.py` (Minnesota)

Each script is annotated and expects you to **wire your data sources**:

- **AB (Canada):**

- **AAFC Annual Crop Inventory (ACI)** (30 m crop type, wheat class; REST/WCS/GEE) [canolacouncil.org], [dairyproducer.com], [agcanada.com]
- **Canadian Drought Monitor (CDM)** monthly classes; **AAFC Drought Watch** indices (SPI/SPEI/soil moisture) [qgis2web.github.io], [docs.qgis.org]
- **AAFC Climate normals** (GDD/CHU) for growing season features [gis4dev.github.io]
- **MODIS MOD16A2 ET** (8-day, 500 m), reprojected to AOI CRS (use `gdal:warpreproject`) [pythongeolab.com], [modis.gsfc.nasa.gov]
- **MN (U.S.):**
 - **NASS CDL** (annual crop mask; wheat) via CroplandCROS [dtnpf.com]
 - **VegScape NDVI** (weekly crop condition) [dtnpf.com]
 - **OpenET ET** (now covers **48 U.S. states**, daily/monthly) via API/UI; export to GeoTIFF [beefweb.com]

In QGIS, use the **Processing Modeler** to chain grid creation, raster sampling, and zonal statistics as you formalize the ETL. (Modeler docs and automation) [app.geo.ca], [open.canada.ca]

Outputs expected (per year):

`features/ab/YYYY_features.npz` and `features/mn/YYYY_features.npz`, with arrays:

- `seq_x` → `[N, T, F_seq]` (weekly NDVI/GDD/ET/stress features)
- `static_x` → `[N, F_static]` (e.g., rotation/topo/drought exposure counts)
- `y` → `[N]` (downscaled yield target; you chose audit as “nice-to-have”—keep it optional)
- `wue` → `[N]` (yield/ET at field/cell scale)
- `group_ids` → `[N]` (province/county id if you want audit)
- `area` → `[N]` (cell area, used for aggregate audit if enabled)

For raster reprojection/resampling and basic raster math (ET deficits, masks), use **GDAL tools** (`gdalwarp`, `gdal_calc`) accessible in QGIS Processing. [modis.gsfc.nasa.gov], [registry.opendata.aws]

2) Train (PyTorch)

Shell

AB (Transformer, no audit)

```
python training/train.py --region ab --years 2019-2025 --model transformer
```

MN with optional audit enabled (if you add `official_totals.json`)

```
python training/train.py --region mn --years 2019-2025 --enable-audit
```

Show more lines

- **Model:** Temporal Transformer (positional encoding + encoder) regresses **yield** and **WUE**.
- **Audit (optional):** If you add `features/<region>/official_totals.json`, the loop applies an **audit loss** for aggregate consistency (province/county). You said auditing is “nice to have,” so it’s opt-in.
- **Dataset:** expects NPZ files from ETL; code infers input dimensions automatically.

If you prefer ConvLSTM later, we can add `convlstm.py` similarly.

3) Publish (QGIS Server & qgis2web)

- **QGIS Server (OGC services)**
Start a QGIS Server instance (Docker or native). Load `qgis-project/ab_wheat_zones.qgs` or `mn_wheat_zones.qgs`. Publish **WMS/WFS/WMTS** using standard `GetCapabilities`, `GetMap`, `GetFeatureInfo` etc. (Server manual & WMS service docs) [etdata.org], [baeri.org]
- **qgis2web (Leaflet)**
In QGIS Desktop, once zones are added, export a **Leaflet** web map using **qgis2web**. Or use our minimal `web/index.html` and point to GeoJSON outputs generated post-training. (Plugin docs & quick guides) [geo.ca], [open.canada.ca]

For **temporal visualization**, QGIS’s **Temporal Controller** (≥ 3.14) is great for weekly animations of NDVI, ET, or evolving zone classes. [farmdocdai...linois.edu]

How zone classes are formed (recommended thresholds)

We’ll compute **three classes** per season using percentiles:

- **High-Potential / High-WUE:** $Yield_pred \geq 70th$ AND $WUE \geq 60th$ → consider higher seeding rate or optimized N timing.
- **High ET / Low-WUE:** $ET_{cum} \geq 70th$ AND $WUE \leq 40th$ → reduce irrigation (if irrigated), emphasize split N timing, consider drought-tolerant varieties.
- **Low Potential / Stable ET:** $Yield_pred \leq 40th$ AND $ET_{cum} < 60th$ → baseline inputs; focus on stand establishment and residue management.

Add **rule constraints** using **CDM exposure weeks (AB)** or **ET anomalies (MN)** to ensure agronomic sense during stress windows. [qgis2web.github.io], [beefweb.com]

Where to customize (quick pointers)

- **ETL wiring (PyQGIS)**

Edit `pyqgis/etl_ab.py` and `pyqgis/etl_mn.py` to bind:

- ACI/CDL crop masks (wheat classes) [\[canolacouncil.org\]](http://canolacouncil.org), [\[dtnpf.com\]](http://dtnpf.com)
- NDVI stacks (CCAP/GEE in AB; VegScape in MN) [\[seedworld.com\]](http://seedworld.com), [\[dtnpf.com\]](http://dtnpf.com)
- ET rasters (MODIS MOD16A2 for AB; OpenET for MN) [\[pythongeolab.com\]](http://pythongeolab.com), [\[beefweb.com\]](http://beefweb.com)
- AAFC climate normals (GDD/CHU for AB) [\[gis4dev.github.io\]](http://gis4dev.github.io)
- CDM monthly drought classes (AB) [\[qgis2web.github.io\]](http://qgis2web.github.io)

- **Feature tensors**

Standardize sequences to weekly timesteps ($T \approx 20\text{--}30$ weeks) and static features to a small set (≤ 30). Save to NPZ with keys expected by `training/utils/dataset.py`.

- **Post-processing (saving outputs)**

In `training/train.py` (TODO at the end), write zone outputs to:

- `outputs/<region>_wheat_zones.gpkg` (QGIS ready)
- `web/data/<region>_zones.geojson` (Leaflet ready)

- **QGIS Server deployment**

Follow the server manual for Docker or native setup and align Desktop/Server versions; publish WMS/WFS endpoints. [\[etdata.org\]](http://etdata.org)

Optional audit (kept minimal, per your preference)

You indicated audit is “nice to have.” If you wish to enable:

- **AB:** Add a small `official_totals.json` using **StatCan** season aggregates (area/production) per sub-region; enable `--enable-audit`. [\[gdal.org\]](http://gdal.org)
- **MN:** Similar approach using **NASS QuickStats** county wheat totals. [\[agr.gc.ca\]](http://agr.gc.ca)

This keeps your model’s **aggregate integrity** without cluttering your workflow.

Why these sources/tools

- **AAFC ACI:** Canada-wide **30 m** crop type maps, high accuracy; ideal for AB wheat field masks. [\[canolacouncil.org\]](http://canolacouncil.org), [\[dairyproducer.com\]](http://dairyproducer.com)

- **CDM & Drought Watch:** Canada's official drought designations + agroclimate indices for stress features. [\[qgis2web.github.io\]](https://qgis2web.github.io/), [\[docs.qgis.org\]](https://docs.qgis.org)
 - **MODIS MOD16A2:** Global ET (8-day, 500 m), reliable for WUE in Canada and easily processed via GDAL. [\[pythongeolab.com\]](https://pythongeolab.com)
 - **CDL & VegScape:** Annual crop masks + NDVI/condition for U.S. grains (MN). [\[dtnpf.com\]](https://dtnpf.com)
 - **OpenET: 48-state** expansion (Dec 16, 2025) for field-scale ET in U.S.—perfect for MN. [\[beefweb.com\]](https://beefweb.com)
 - **QGIS Processing Modeler/DB Manager:** repeatable ETL chains; robust PostGIS integration. [\[app.geo.ca\]](https://app.geo.ca), [\[canadiangeo.ca/graphics\]](https://canadiangeo.ca/graphics)
 - **QGIS Server & qgis2web:** publish OGC services or quick Leaflet maps straight from your QGIS projects. [\[etdata.org\]](https://etdata.org), [\[geo.ca\]](https://geo.ca)
-

Next steps I can do for you

1. **Wire the ETL to real endpoints** (ACI WCS, CDM layers, VegScape API, OpenET downloads) and produce first NPZ features for **AB 2023 + MN 2023**.
2. **Add post-processing** in `train.py` to auto-save zone GeoPackage/GeoJSON after inference.
3. **Set up QGIS Server (Docker)** with the included projects and give you **sample WMS/WFS URLs**.
4. **Export a Leaflet map** via `qgis2web` with your zone symbology and pop-ups.

If you share any preferred endpoints/tokens (e.g., OpenET API auth) or an AOI boundary file for AB/MN, I'll plug them in and run the first pipeline pass.

Food Security Index

A clean way to describe **food security** with math is to formalize the four FAO dimensions—**Availability, Access, Utilization, Stability**—with variables, constraints, and indices.

Composite Food Security Index (FSI)

To avoid one pillar masking another, use a **geometric mean** (penalizes low scores):

$$FSI_{i,t} = (A_{i,t}^{w_A} \cdot Access_{i,t}^{w_X} \cdot U_{i,t}^{w_U} \cdot S_{i,t}^{w_S})^{1/w_A + w_X + w_U + w_S}$$

where $U_{i,t}$ is a market-level utilization index (e.g., average of $U_{h,t}$ across households), and w_h are weights.

If you want **trustable, regularly updated data for the FAO's four food security dimensions—Availability, Access, Utilization, Stability**—the safest route is to pull from FAO's official portals and the UN-custodian datasets they manage or endorse. Below is a practical map of **what to use for each pillar, where to download it, and how to validate**.

The most reliable sources (by pillar)

1) Availability (supply at national/market level)

- **FAOSTAT – Food Balance Sheets (FBS)**: shows supply–utilization of each commodity, including **dietary energy, protein, and fat per capita**. This is the canonical source to quantify national calorie and macronutrient availability. [[fao.org](https://www.fao.org/)], [[index.nut...tufts.edu](https://index.nut.tufts.edu/)]
- **FAOSTAT Portal (main entry)**: browse/download FBS, Production, Trade domains; same data are accessible via API and bulk CSV. [[repository...arvard.edu](https://repository.harvard.edu/)], [en.wikipedia.org]
- **New FAOSTAT “Food & Diet” domain**: harmonized food and nutrient statistics across supply and intake (2010+), useful to link supply (FBS) with apparent consumption. [accessagric.com]

What to download: FBS time series (2010–2022+), plus Production and Trade for cross-checks. [[fao.org](https://www.fao.org/)], [[repository...arvard.edu](https://repository.harvard.edu/)]

2) Access (affordability/price access)

- **SDG 2.1.2 — FIES** (Food Insecurity Experience Scale): official FAO indicator for **moderate or severe food insecurity**, based on people's experiences (Voices of the Hungry). You can get microdata documentation and tools; country microdata are often archived via the World Bank Microdata Library. [[fao.org](https://www.fao.org/)], [[microdata...lbank.org](https://microdata.worldbank.org/)]
- **Cost & Affordability of a Healthy Diet (CoAHD) — FAOSTAT**: global indicators on the **cost of a healthy diet** and **prevalence of unaffordability**, produced by FAO with the World Bank's Food Prices for Nutrition project. Downloadable datasets + methods briefs. [data.apps.fao.org], [[openknowle...ge.fao.org](https://openknowledge.fao.org/)]
- **FPMA (Food Price Monitoring & Analysis)**: **domestic monthly retail/wholesale prices** for staple foods (~100+ countries), with tools to analyze anomalies and seasonality—critical for affordability monitoring at market level. [[fao.org](https://www.fao.org/)], [[fao.org](https://www.fao.org/)]

What to download: CoAHD indicators (country/year), FIES (SDG 2.1.2) series, and FPMA domestic prices for commodities you track. [\[data.apps.fao.org\]](https://data.apps.fao.org), [\[fao.org\]](https://fao.org), [\[fao.org\]](https://fao.org)

3) Utilization (diet quality & nutrition outcomes)

- **FAO/UNICEF/WHO/World Bank Joint Child Malnutrition Estimates (JME):** global, regional, and country **stunting, wasting, overweight** trends (2000–2024), widely used for utilization outcomes. [\[data.unicef.org\]](https://data.unicef.org), [\[who.int\]](https://who.int)
- **FAOSTAT “Food & Diet” domain:** adds harmonized **nutrient intake/availability statistics**, connecting supply to apparent/individual consumption where surveys exist. [\[accessagric.com\]](https://accessagric.com)

What to download: JME CSVs for child nutrition indicators, plus FAOSTAT nutrient availability/intake if available for your countries. [\[data.unicef.org\]](https://data.unicef.org), [\[accessagric.com\]](https://accessagric.com)

4) Stability (risk/volatility over time)

- **FPMA + FPMA Bulletins:** track **domestic price volatility and anomalies** (SDG 2.c.1), with international benchmark series; ideal for short-term stability signals. [\[fao.org\]](https://fao.org), [\[fpma.fao.org\]](https://fpma.fao.org)
- **GIEWS (Global Information and Early Warning System):** authoritative **early warnings, country briefs, crop prospects, and special alerts**, with cereal balance sheets and earth observation tools (ASIS). [\[fao.org\]](https://fao.org), [\[fao.org\]](https://fao.org)

What to download: FPMA country/commodity price series; GIEWS **Crop Prospects & Food Situation** reports and Country Cereal Balance Sheets for seasonal supply–demand risks. [\[fao.org\]](https://fao.org), [\[fao.org\]](https://fao.org)

Where to click & how to retrieve

1. **FAOSTAT portal** → **Data** → choose domain (e.g., **Food Balances, Food Security & Nutrition, Food & Diet, Production, Trade**). Export as CSV or use the built-in “Compare/Visualize” tools. [\[repository...arvard.edu\]](https://repository...arvard.edu)
2. **SDG Data Portal (FAO custodianship)** → **Indicator 2.1.1 (PoU)** and **2.1.2 (FIES)** pages have **download links** and metadata (methods, periodicity). [\[fao.org\]](https://fao.org), [\[unstats.un.org\]](https://unstats.un.org)
3. **CoAHD (Cost & Affordability of a Healthy Diet)** → FAOSTAT dataset page provides **schema, SQL query file**, and **bulk downloads**; method brief explains computations and update cadence. [\[data.apps.fao.org\]](https://data.apps.fao.org), [\[data.apps.fao.org\]](https://data.apps.fao.org), [\[data.apps.fao.org\]](https://data.apps.fao.org)

4. **FPMA Tool** → browse **domestic price series** (retail & wholesale) and **international benchmarks**; export series, check anomaly counts (SDG 2.c.1). [fpma.fao.org]
 5. **GIEWS** → fetch **FPMA bulletins**, **Country Briefs**, **Crop Prospects & Food Situation**, and **Special Alerts**; useful for contextual stability assessment. [fao.org]
 6. **JME datasets** → use UNICEF/WHO/World Bank download pages and interactive dashboards (CSV/xlsx). [data.unicef.org], [[datatopics...lbank.org](https://datatopics.worldbank.org)]
 7. **Hand-in-Hand Geospatial Platform** → combine FAOSTAT with geospatial layers (water stress, agro-ecology, prices) for **spatial analysis** and visualization. [fao.org], [data.apps.fao.org]
-

Quick mapping: FAO dimensions → indicators you can operationalize

- **Availability:**
 - *Dietary Energy Supply (kcal/cap/day), protein/fat supply* from **FBS** (national level). [fao.org]
 - *Commodity balances, production & trade* from **FAOSTAT** domains. [[repository...arvard.edu](https://repository.harvard.edu)]
 - **Access:**
 - *Affordability of a healthy diet (PUA, NUA, cost per person/day)* from **CoAHD**. [data.apps.fao.org]
 - *Moderate/severe food insecurity prevalence* from **FIES (SDG 2.1.2)**. [fao.org]
 - *Market access stress* via **FPMA domestic prices** and anomaly counts (SDG 2.c.1). [fao.org]
 - **Utilization:**
 - *Stunting, wasting, overweight* from **JME (UNICEF/WHO/WB)**. [who.int]
 - *Nutrient intake/availability* via FAOSTAT **Food & Diet** domain where available. [accessagric.com]
 - **Stability:**
 - *Price volatility/anomalies* from **FPMA**. [fpma.fao.org]
 - *Seasonal supply risk, early warnings* from **GIEWS** reports & country cereal balance sheets. [fao.org]
-

Validation & “trustability” checklist (use this to keep your data clean)

- **Use custodial indicators for SDGs:** FAO is the **custodian agency** for SDG 2.1.1 (PoU) and 2.1.2 (FIES); always pull PoU from FAO's SDG portal and not secondary blogs. [\[unstats.un.org\]](https://unstats.un.org)
 - **Cross-check supply with FBS:** PoU uses **dietary energy consumption and MDER** derived from FBS; ensure your availability trends match PoU movements. [\[angela-smi....github.io\]](https://angela-smi.github.io)
 - **Document methodology:** For affordability, cite FAO/World Bank **CoAHD methods** (Healthy Diet Basket, PPP adjustments), and keep versions aligned with latest **SOFI** updates. [\[openknowle...ge.fao.org\]](https://openknowledge.fao.org)
 - **Triangulate utilization:** Pair FAOSTAT nutrient statistics with **JME outcomes** to avoid mistaking supply for actual nutrition status. [\[accessagric.com\]](https://accessagric.com), [\[data.unicef.org\]](https://data.unicef.org)
 - **Monitor current events:** For stability, combine **FPMA price series** with **GIEWS alerts**—this catches shocks that don't yet show in annual datasets. [\[fao.org\]](https://fao.org), [\[fao.org\]](https://fao.org)
-

Minimal starter list: what to download for one country

1. **FBS (FAOSTAT)** → dietary energy, protein, fat per capita; commodity supply/utilization. [\[fao.org\]](https://fao.org)
2. **SDG 2.1.1 (PoU)** → hunger prevalence + metadata. [\[fao.org\]](https://fao.org)
3. **SDG 2.1.2 (FIES)** → moderate/severe food insecurity (and survey microdata where available). [\[microdata....ldb.org\]](https://microdata.ldb.org)
4. **CoAHD (FAOSTAT)** → cost of healthy diet, prevalence & number unable to afford it. [\[data.apps.fao.org\]](https://data.apps.fao.org)
5. **FPMA** → monthly domestic price series for staples. [\[fpma.fao.org\]](https://fpma.fao.org)
6. **GIEWS** → latest country brief, crop prospects, and any special alerts. [\[fao.org\]](https://fao.org)
7. **JME** → stunting, wasting, overweight CSVs. [\[data.unicef.org\]](https://data.unicef.org)

model

We used a **gradient-boosted tree model** to explain how climate and disaster shocks drive year-to-year changes in food production, which is the main pathway to food insecurity.

what model was used:

A gradient-boosted decision tree model
(specifically: *HistGradientBoostingRegressor*)

In human terms:

- It's a smart stack of many small **if-then rules**
- Each new rule fixes mistakes from the last ones
- It's good at **messy, real-world data** (weather, disasters)

Think:

"If it was hotter AND there were more disasters, food usually dropped... unless the country is rich."

what did it do:

It predicted change, not totals.

For each country and year, it learned:

"Compared to last year, did food production go **up or down**, and by how much?"

To do that, it looked at:

- temperature
- rainfall
- disasters
- economic strength

- last year's conditions

It **did NOT**:

- predict total food supply
- predict hunger directly
- compare rich vs poor countries unfairly

It only learned **cause** → **reaction** patterns.

why is this important:

Because this matches how the real world works.

Climate doesn't suddenly decide:

"Nigeria shall produce 85 food units this year."

Climate causes **shocks**:

- heat waves
- floods
- droughts

Those shocks cause:

- **drops or bumps** in food production

And *repeated drops* lead to:

- higher food prices
- reliance on imports
- food insecurity

So your model:

- isolates the **first impact**
- avoids political/policy noise
- shows a clear pathway from climate → food insecurity

sensitivity results (test)

Dropped 5 rows with missing change target. Remaining: 110

In-sample diagnostics (change model)

MAE: 0.9909744774794419

R² : 0.9014464098702681

Sensitivity analysis (mean effect on predicted food production change):

Scenario	Baseline mean prediction	Scenario mean prediction	Effect (scenario - baseline)
-10% precipitation	1.746545	1.347499	-0.399046
+1°C temperature	1.746545	1.531614	-0.214931
-10% precipitation (lagged only)	1.746545	1.665628	-0.080917
+30% disasters (lagged only)	1.746545	1.673929	-0.072616
+30% disasters	1.746545	1.834335	0.087789
+1°C temperature (lagged only)	1.746545	1.949582	0.203036

We changed one climate factor at a time and watched how food production growth reacted. Less rain and higher temperatures reduce production right away. Effects from previous years are smaller because countries adapt and recover.

USING ALL DATA:

Dropped 5 rows with missing change target. Remaining: 110

In-sample diagnostics (change model)

MAE: 0.748946660370453

R² : 0.944045748065334

Sensitivity analysis (mean effect on predicted food production change):

Scenario	Baseline mean prediction	Scenario mean prediction	Effect (scenario - baseline)
-10% precipitation	1.746545	1.357884	-0.388662
+30% disasters (lagged only)	1.746545	1.691492	-0.055053
-10% precipitation (lagged only)	1.746545	1.702001	-0.044544
+30% disasters	1.746545	1.764431	0.017886
+1°C temperature	1.746545	1.778082	0.031537
+1°C temperature (lagged only)	1.746545	2.147390	0.400844

presentation

1. Analogy to something else
 - a. Tons of problems require math, but as useful as long division is, you're not going to do long strings of mental math with say, corporate tax returns. Of course, you're going to reach for a calculator instead.
2. Problem - same thing w/ analogy, need more accessible quick predictors that are still accurate.
 - a. Food security is great and everything, but plenty of places have insecurity, including rich countries.
 - b. Lots of great ideas for how to solve food insecurity, but oftentimes rely on being able to predict
 - c. Policy relies on if we have quickly accessible and reliable predictors
 - i. Need to have large long term consistently tracked factors
3. Solution
 - a. As so happens, agricultural output, as well as the prices of those commodities are one of the most accessible and consistently tracked
 - b. Building a predictive model off of these factors, test if works then idea is very exciting
4. How research does that
 - a. Used data from (sources), aggregated Dietary energy, protein supply, GDP per capita, and food production index to create a Food Security Index (FSI), using 1970-2012 and testing it on 2013-2023
 - i. Take in data on corn, soybean, and wheat yield production and harvest in US
 - b. Used a Random Forest model to train an AI, takes this info and trains it to look for patterns. If these patterns are connected to food security, then this data can be used to quickly and accessibly predict national levels of food security
 - i. Tested other models but Random Forest plus shocks and lags came out the most accurate, also reduced risk of overfitting (first model was 95%)
 - ii. Importantly, we tried to **predict change in the FSI index rather than absolute FSI values**
 1. This is due to FSI being very persistent
 - c. Shocks and lags help improve accuracy
 - i. Lags go back 2 years, determine if something affects the index long term
 - d. Performance metrics
 - i. (RMSE/MAE/R², YoY improvement/declines)
5. Where applicable
 - a. In policy
 - b. Knowing when a crisis may occur
 - c. Year to year changes
6. Impacts of that

Hook:

-

Context:

- Although it is one of the wealthiest countries in the world, the United States still faces food insecurity.

Research:

Impact: which ones contribute the most to food insecurity

Closing:

questions/terms and stuff :D

Random Forest Model

A **Random Forest** is an ensemble machine-learning method that builds **many decision trees** and then combines their predictions.

- Each tree sees a *random subset* of the data and features
- The forest then **averages** (for regression) or **votes** (for classification)
- This reduces overfitting and usually improves accuracy and stability

Analogy:

Instead of trusting one person's opinion (one tree), you ask 200 people (the forest) and take the consensus.

Regression Modelling

Regression models are used when you want to **predict a continuous numerical value**.

Examples:

- Predicting house prices
- Estimating temperature
- Forecasting sales

Common regression types:

- **Linear regression** (straight-line relationship)
- **Polynomial regression**
- **Random forest regression**
- **Neural network regression**

The goal is to learn a function $f(x)$ that predicts a numeric output y .

R² Value (Coefficient of Determination)

R² measures **how much of the variance in the target variable your model explains**.

- Ranges from **0 to 1**
- **1.0** → perfect prediction
- **0.0** → model explains nothing
- Can be **negative** if the model is worse than predicting the mean

Interpretation example:

$R^2 = 0.82$ → "Your model explains 82% of the variation in the data."

MAE (Mean Absolute Error)

MAE measures the **average absolute difference** between predictions and actual values.

$$\mathrm{MAE} = \frac{1}{n} \sum |y - \hat{y}|$$

- Lower MAE = better
- More robust to outliers than RMSE
- Also in the same units as the target

Use MAE when:

You want a simple, intuitive measure of average error.